

Functional Analysis HW

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§1 HW1

§1.1 Problem 1

From P2. of Preliminaries.

Theorem 1.1

Proposition 6. Let E be a finite dimensional vector space. Then every two norms on E are equivalent.

We assume that vector space X is finite dimensional with $\dim X = n$. Then, $\forall x \in X$, we have $x = \xi_1 e_1 + \cdots + \xi_n e_n$.

In the sense of algebraic isomorphism, **2 finite dimension vector spaces are isomorphic. $\iff$ They have the same dimension.**

Now, we consider their relationship in the sense of Topology. $\forall x \in X$, we have x corresponds to a point $\xi = Tx \triangleq (\xi_1, \dots, \xi_n)$ in space $\mathbb{K}^n$. Naturally, we hope to establish a continuous mapping between x 's norm in X and $\xi = Tx$'s norm

$$\|\xi\| = \|Tx\| := \left(\sum_{j=1}^n |\xi_j|^2 \right)^{\frac{1}{2}}$$

in $\mathbb{K}^n$. So, we consider the mapping

$$p(\xi) := \left\| \sum_{j=1}^n \xi_j e_j \right\| \quad (\forall \xi \in \mathbb{K}^n)$$

Proposition 1.2

p is uniformly continuous for ξ .

Proof. Factually, $\forall \xi = (\xi_1, \dots, \xi_n), \eta = (\eta_1, \dots, \eta_n) \in \mathbb{K}^n$, we have:

$$\begin{aligned} p(\xi) - p(\eta) &\leq p(\xi - \eta) \leq \sum_{i=1}^n |\xi_i - \eta_i| \cdot \|e_i\| \\ &\leq \left(\sum_{i=1}^n \|\xi_i - \eta_i\|^2 \right)^{\frac{1}{2}} \cdot \left(\sum_{i=1}^n \|e_i\|^2 \right)^{\frac{1}{2}} \leq \|\xi - \eta\| \cdot \left(\sum_{i=1}^n \|e_i\|^2 \right)^{\frac{1}{2}} \end{aligned}$$

□

Proposition 1.3**Positive homogeneity of p .**

Proof. By the **Homogeneity of Norms**, $\forall \xi \in \mathbb{K}^n - \{\mathbf{0}\}$, we have:

$$p(\xi) = |\xi| \left\| \sum_{j=1}^n \frac{|\xi_j|}{|\xi|} e_j \right\| = |\xi| \cdot p\left(\frac{\xi}{|\xi|}\right) \quad (1)$$

□

Proof. Now, we prove the **equivalence of norms**. We only need to find some constant $C_1, C_2 > 0$.

We notice that $S = \{\xi \in \mathbb{K}^n : |\xi| = 1\}$ is compact. So, p attains its bounds on S . $\exists \text{ constant } C_1, C_2 \geq 0$, we have:

$$0 \leq C_1 \leq p(\xi) \leq C_2, \forall \xi \in S$$

By the **Positive homogeneity** of p , $\forall \xi \in \mathbb{K}^n - \{\mathbf{0}\}$, we have:

$$C_1|\xi| \leq p(\xi) \leq C_2|\xi|$$

Now we prove that $C_1 > 0$. If not, $\exists \xi_0 \in S$ such that $p(\xi_0) = 0$. So, $\sum_{j=1}^n \xi_{0,j} e_j = \mathbf{0}$. Since $\{e_1, \dots, e_n\}$ is a basis of X , we have $\xi_0 = \mathbf{0}$, which contradicts the fact that $\xi_0 \in S$.

Finally, $\forall x \in X - \{\mathbf{0}\}$, we have:

$$0 < C_1|Tx| \leq \|x\| \leq C_2|Tx|$$

If we view $|Tx|$ as another norm on X , i.e. $\|x\|_T := |Tx|$, then the proof is complete. □

Remark 1.4. $|\xi| = (\sum_{j=1}^n |\xi_j|^2)^{\frac{1}{2}}$ and $|\xi| = \sum_{j=1}^n |\xi_j|$ are both ok, because we aim to prove $\forall$ norm on this n.v.s is equivalent to $|\xi|$.

§1.2 Problem 2

From P2. of Preliminaries.

Theorem 1.5

Proposition 9. Let E and F be two normed vector spaces, and A is a linear operator from E to F . Then A is bounded $\iff A$ is continuous.

Proof. " $\implies$ " is trivial.

" $\impliedby$ ": We prove by contradiction. If A is not bounded, then $\forall n \in \mathbb{N}^+$, $\exists x_n \in E$ such that $\|x_n\| \leq 1$ and $\|Ax_n\| > n$. So, we have $\frac{\|Ax_n\|}{\|x_n\|} > n$.

i.e.

$$\left\| A \left(\frac{x_n}{n \cdot \|x_n\|} \right) \right\| > 1$$

Let $y_n = \frac{x_n}{n \cdot \|x_n\|}$, then we have $\|A(y_n)\| > 1$, but $y_n \rightarrow \mathbf{0} (n \rightarrow \infty)$, which contradicts the continuity of A at $\mathbf{0}$. □

Remark 1.6. From Tutorial 1 Proposition A is continuous at a point $x_0 \in n.v.s.X \implies A$ is bounded.

Proof. A is continuous at $x_0 \iff A$ is continuous at 0. [Translation is continuous.]
 $\forall \epsilon > 0, \exists \delta > 0$ such that $\|x\| < \delta \implies \|Ax\| < \epsilon$. We take $\epsilon = 1$, let $y = \frac{\delta}{2} \frac{x}{\|x\|} \in B(0, \delta)$.
 Then we have $\|A(y)\| < 1 \implies \frac{\delta}{2\|x\|} \cdot \|Ax\| < 1 \implies \|Ax\| < \frac{2}{\delta} \|x\|$. Hence A is bounded. $\square$

§1.3 Problem 3

Theorem 1.7

1.2 Let E be a vector space of dimension n and let $(e_i)_{1 \leq i \leq n}$ be a basis of E . Given $x \in E$, write $x = \sum_{i=1}^n x_i e_i$ with $x_i \in \mathbb{R}$; given $f \in E^*$, set $f_i = \langle f, e_i \rangle$. 1. Consider on E the norm

$$\|x\|_1 = \sum_{i=1}^n |x_i|.$$

(a) Compute explicitly, in terms of the f_i 's, the dual norm $\|f\|_{E^*}$ of $f \in E^*$.

(b) Determine explicitly the set $F(x)$ (duality map) for every $x \in E$.

2. Same questions but where E is provided with the norm

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

3. Same questions but where E is provided with the norm

$$\|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2},$$

and more generally with the norm

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad \text{where } p \in (1, \infty).$$

Proof. 1(a). We want to prove that

$$\|f\|_{E^*} = \max_{1 \leq i \leq n} |f_i|.$$

By definition,

$$\|f\|_{E^*} = \sup_{x \neq 0} \frac{|\langle f, x \rangle|}{\|x\|_1} = \sup_{x \neq 0} \frac{\langle f, \sum_{i=1}^n x_i e_i \rangle}{\|x\|_1} = \sup_{x \neq 0} \frac{\sum_{i=1}^n \langle f, e_i \rangle x_i}{\|x\|_1} = \sup_{x \neq 0} \frac{\sum_{i=1}^n \langle f, e_i \rangle}{\sum_{i=1}^n |x_i|}$$

We assume that $\sum_{i=1}^n |x_i| = 1$, then

$$\|f\|_{E^*} = \sup_{\|x\|_1=1} \sum_{i=1}^n f_i x_i \leq \max_{1 \leq i \leq n} |f_i| \cdot \sup_{\|x\|_1=1} \sum_{i=1}^n |x_i| = \max_{1 \leq i \leq n} |f_i| \cdot 1 = \max_{1 \leq i \leq n} |f_i|$$

On the other hand, let k be such that $|f_k| = \max_{1 \leq i \leq n} |f_i|$. If we take x such that $x_k = \text{sign } f_k$ and $x_i = 0$ for $i \neq k$, then $\|x\|_1 = 1$ and $\langle f, x \rangle = |f_k|$. It follows that

$$\|f\|_{E^*} \geq |f_k| = \max_{1 \leq i \leq n} |f_i|.$$

1(b). Let's prove $f \in F(x)$ iff for every $1 \leq i \leq n$ one has

$$f_i = \begin{cases} (\text{sign } x_i) \|x\|_1 & \text{if } x_i \neq 0, \\ \text{anything in the interval } [-\|x\|_1, +\|x\|_1] & \text{if } x_i = 0. \end{cases}$$

Then, by definition, $\|f\| = \sup_{x \neq 0} \frac{\langle f, x \rangle}{\|x\|} = \sup_{x \neq 0} \frac{\sum x_i f_i}{\|x\|}$.

Use the same method in 1(a), we have $\|f\| \leq \max |f_i|$. If f satisfies the condition above, then $\|f\| = \max |f_i| = \|x\|_1$. And $\langle f, x \rangle = \sum x_i f_i = \sum |x_i| \|x\|_1 = \|x\|_1^2$. So, $\frac{\langle f, x \rangle}{\|x\|} = \|x\|_1 = \|f\|$. Hence $f \in F(x)$.

2(a). We want to prove that

$$\|f\|_{E^*} = \sum_{i=1}^n |f_i|.$$

By definition,

$$\begin{aligned} \|f\|_{E^*} &= \sup_{x \neq 0} \frac{|\langle f, x \rangle|}{\|x\|_\infty} = \sup_{x \neq 0} \frac{\sum_{i=1}^n f_i x_i}{\max_{1 \leq i \leq n} |x_i|} \\ &\leq \frac{\max_{1 \leq i \leq n} |x_i| \cdot \sum_{i=1}^n |f_i|}{\max_{1 \leq i \leq n} |x_i|} = \sum_{i=1}^n |f_i|. \end{aligned}$$

On the other hand, let x be such that $x_i = \text{sign } f_i$. Then $\|x\|_\infty = 1$ and $\langle f, x \rangle = \sum_{i=1}^n |f_i|$. It follows that

$$\|f\|_{E^*} \geq \sum_{i=1}^n |f_i|.$$

2(b). Given $x \in E$ consider the set

$$I = \{1 \leq i \leq n; |x_i| = \|x\|_\infty\}.$$

Then $f \in F(x)$ iff one has (i) $f_i = 0 \ \forall i \notin I$, (ii) $f_i x_i \geq 0 \ \forall i \in I$ and $\sum_{i \in I} |f_i| = \|x\|_\infty$.

Let's prove the condition for $f \in F(x)$. By definition, $f \in F(x)$ iff

$$\|f\|_{E^*} = \frac{|\langle f, x \rangle|}{\|x\|_\infty}$$

We have proved that $\|f\|_{E^*} = \sum_{i=1}^n |f_i|$. So, $f \in F(x)$ iff

$$\sum_{i=1}^n |f_i| = \frac{|\sum_{i=1}^n f_i x_i|}{\|x\|_\infty}.$$

By definition of I , we have

$$\sum_{i=1}^n |f_i| = \sum_{i \in I} |f_i| + \sum_{i \notin I} |f_i| \geq \frac{|\sum_{i \in I} f_i x_i|}{\|x\|_\infty} + \sum_{i \notin I} |f_i|.$$

It follows that $f \in F(x)$ implies $\sum_{i \notin I} |f_i| = 0$ and $\sum_{i \in I} |f_i| = \frac{|\sum_{i \in I} f_i x_i|}{\|x\|_\infty}$. The first condition is (i). For the second condition, we have

$$\sum_{i \in I} |f_i| \geq \frac{|\sum_{i \in I} f_i x_i|}{\|x\|_\infty} \geq \frac{\sum_{i \in I} |f_i| |x_i|}{\|x\|_\infty} = \sum_{i \in I} |f_i|.$$

It follows that $f_i x_i \geq 0$ for all $i \in I$ and $\sum_{i \in I} |f_i| = \|x\|_\infty$. This is (ii). Conversely, if f satisfies (i) and (ii), then

$$\sum_{i=1}^n |f_i| = \sum_{i \in I} |f_i| = \|x\|_\infty = \frac{|\sum_{i \in I} f_i x_i|}{\|x\|_\infty} = \frac{|\sum_{i=1}^n f_i x_i|}{\|x\|_\infty}.$$

Hence $f \in F(x)$.

3(a). We want to prove that

$$\|f\|_{E^*} = \left(\sum_{i=1}^n |f_i|^2 \right)^{1/2}$$

and $f \in F(x)$ iff one has $f_i = x_i \ \forall i = 1, 2, \dots, n$. More generally,

$$\|f\|_{E^*} = \left(\sum_{i=1}^n |f_i|^q \right)^{1/q},$$

where $\frac{1}{p} + \frac{1}{q} = 1$, and $f \in F(x)$ iff one has $f_i = |x_i|^{p-2} \frac{x_i}{\|x\|_p^{p-2}}, \forall i = 1, 2, \dots, n$.

Now, we prove the general case. By definition,

$$\|f\|_{E^*} = \sup_{x \neq 0} \frac{|\langle f, x \rangle|}{\|x\|_p} = \sup_{x \neq 0} \frac{\sum_{i=1}^n f_i x_i}{(\sum_{i=1}^n |x_i|^p)^{1/p}}$$

By Hölder's inequality, we have

$$\|f\|_{E^*} \leq \sup_{x \neq 0} \frac{(\sum_{i=1}^n |f_i|^q)^{1/q} \cdot (\sum_{i=1}^n |x_i|^p)^{1/p}}{(\sum_{i=1}^n |x_i|^p)^{1/p}} = (\sum_{i=1}^n |f_i|^q)^{1/q}$$

On the other hand, let x be such that $x_i = \text{sign } f_i \cdot |f_i|^{q-1}$. Then $\|x\|_p = (\sum_{i=1}^n |f_i|^q)^{1/p}$ and $\langle f, x \rangle = \sum_{i=1}^n |f_i|^q$. It follows that

$$\|f\|_{E^*} \geq \frac{\sum_{i=1}^n |f_i|^q}{(\sum_{i=1}^n |f_i|^q)^{1/p}} = (\sum_{i=1}^n |f_i|^q)^{1/q}.$$

Thus, we have proved that

$$\|f\|_{E^*} = (\sum_{i=1}^n |f_i|^q)^{1/q}.$$

3(b). Now, we prove the condition for $f \in F(x)$. By definition, $f \in F(x)$ iff

$$\|f\|_{E^*} = \frac{|\langle f, x \rangle|}{\|x\|_p}$$

Let's prove the general case. We have proved that $\|f\|_{E^*} = (\sum_{i=1}^n |f_i|^q)^{1/q}$. So, $f \in F(x)$ iff

$$(\sum_{i=1}^n |f_i|^q) = \frac{|\sum_{i=1}^n f_i x_i|}{\|x\|_p}.$$

By Hölder's inequality, we have

$$(\sum_{i=1}^n |f_i|^q)^{1/q} \geq \frac{|\sum_{i=1}^n f_i x_i|}{\|x\|_p} \leq (\sum_{i=1}^n |f_i|^q)^{1/q}.$$

It follows that $f \in F(x)$ implies that the equality holds in Hölder's inequality, which is equivalent to the existence of $\lambda \geq 0$ such that

$$|f_i|^q = \lambda |x_i|^p, \forall i = 1, 2, \dots, n.$$

We have

$$\sum_{i=1}^n |f_i|^q = \lambda \sum_{i=1}^n |x_i|^p.$$

Thus, we conclude that

$$\|f\|_{E^*} = \frac{|\langle f, x \rangle|}{\|x\|_p}.$$

$\iff \lambda^{1/q} = \frac{|\langle f, x \rangle|}{\|x\|_p^{p/q}}$. It follows that

$$|f_i|^q = \frac{|\langle f, x \rangle|^q}{\|x\|_p^p} |x_i|^p, \forall i = 1, 2, \dots, n.$$

Using the fact that $\langle f, x \rangle = \sum_{i=1}^n f_i x_i$, we conclude that

$$f_i = |x_i|^{p-2} \frac{x_i}{\|x\|_p^{p-2}}, \forall i = 1, 2, \dots, n.$$

Thus, we have proved that $f \in F(x)$ iff one has $f_i = |x_i|^{p-2} \frac{x_i}{\|x\|_p^{p-2}}, \forall i = 1, 2, \dots, n$. □

§1.4 Problem 4

Theorem 1.8

[1.5] Let E be an infinite-dimensional n.v.s.

1. Prove (using **Zorn's lemma**) that there exists an algebraic basis $(e_i)_{i \in I}$ in E such that $\|e_i\| = 1, \forall i \in I$.

Recall that an algebraic basis (or **Hamel basis**) is a subset $(e_i)_{i \in I}$ in E such that every $x \in E$ may be written uniquely as

$$x = \sum_{i \in J} x_i e_i \text{ with } J \subset I, J \text{ finite.}$$

2. Construct a linear functional $f : E \rightarrow \mathbb{R}$ that is not continuous.

3*. Assuming in addition that E is a Banach space, prove that I is not countable. [Hint: Use Baire category theorem (Theorem 2.1).]

Proof. 1. Let E be an infinite-dimensional normed vector space. We aim to construct an algebraic basis $(e_i)_{i \in I}$ such that $\|e_i\| = 1$ for all $i \in I$.

Define the collection:

$$\mathcal{S} = \{A \subseteq E \mid A \text{ is linearly independent and } \|a\| = 1 \text{ for all } a \in A\}.$$

Since E is infinite-dimensional, there exists a nonzero vector $v \in E$; then $v/\|v\| \in \mathcal{S}$, so $\mathcal{S}$ is nonempty.

Partially order $\mathcal{S}$ by inclusion. Let $\{A_\alpha\}$ be a chain in $\mathcal{S}$. Define:

$$A = \bigcup_{\alpha} A_{\alpha}.$$

Any finite subset of A is contained in some A_α , hence linearly independent. Also, every $a \in A$ satisfies $\|a\| = 1$. Thus, $A \in \mathcal{S}$ is an upper bound for the chain.

By **Zorn's Lemma**, $\mathcal{S}$ has a maximal element B . We claim B is an algebraic basis for E .

Suppose there exists $x \in E$ not in the span of B . Then $v = x/\|x\|$ satisfies $\|v\| = 1$. If v were linearly dependent on B , then x would be in the span of B , a contradiction. Hence, $B \cup \{v\}$ is linearly independent, contradicting the maximality of B .

Therefore, B is an algebraic basis. Denote it by $(e_i)_{i \in I}$, with $\|e_i\| = 1$ for all $i \in I$.

2. Construction of a Discontinuous Linear Functional

Since E is infinite-dimensional, the index set I is infinite. Choose a countable subset $I_0 \subseteq I$, say $I_0 = \{1, 2, 3, \dots\}$, and define a linear functional $f : E \rightarrow \mathbb{R}$ as follows:

- For $i \in I_0$, set $f(e_i) = i$,
- For $i \in I \setminus I_0$, set $f(e_i) = 0$.

Every $x \in E$ has a unique representation:

$$x = \sum_{i \in J} x_i e_i, \quad \text{with } J \subset I \text{ finite.}$$

Define:

$$f(x) = \sum_{i \in J} x_i f(e_i).$$

This defines a linear functional on E .

However, f is **not continuous**. For each $n \in \mathbb{N}$, we have $\|e_n\| = 1$, but $|f(e_n)| = n$. If f were continuous, there would exist $M > 0$ such that $|f(x)| \leq M\|x\|$ for all $x \in E$, which is contradicted by $f(e_n) = n \rightarrow \infty$.

3*. Uncountability of the Basis in a Banach Space

Assume now that E is a **Banach space**. Suppose, for contradiction, that the index set I is countable.

Every $x \in E$ can be written uniquely as:

$$x = \sum_{i \in J} x_i e_i, \quad \text{with } J \subset I \text{ finite.}$$

Thus:

$$E = \bigcup_{\substack{J \subset I \\ J \text{ finite}}} \text{span}\{e_j : j \in J\}.$$

Since I is countable, the set of all finite subsets $J \subset I$ is also countable. Each $\text{span}\{e_j : j \in J\}$ is a finite-dimensional subspace, hence **closed**. Moreover, since E is infinite-dimensional, each such subspace is **proper** and has **empty interior**.

Therefore, E is a countable union of nowhere dense sets, contradicting the **Baire Category Theorem** (a complete metric space cannot be a countable union of nowhere dense sets).

Hence, I must be **uncountable**.

□

Remark 1.9. Solution on book P373. 1. Let $\mathcal{P}$ denote the family of all linearly independent subsets of E . It is easy to see that $\mathcal{P}$ (ordered by the usual inclusion) is inductive. Zorn's lemma implies that $\mathcal{P}$ has a maximal element, denoted by $(e_i)_{i \in I}$, which is clearly an algebraic basis. Since $e_i \neq 0 \forall i \in I$, one may assume, by normalization, that $\|e_i\| = 1 \forall i \in I$.

2. Since E is infinite-dimensional one may assume that $\mathbb{N} \subset I$. There exists a (unique) linear functional on E such that $f(e_i) = i$ if $i \in \mathbb{N}$ and $f(e_i) = 0$ if $i \in I \setminus \mathbb{N}$.

3. Assume that I is countable, i.e., $I = \mathbb{N}$. Consider the vector space F_n spanned by $(e_i)_{i \leq n}$. F_n is closed (see Section 11.1) and, moreover, $\bigcup_{n=1}^{\infty} F_n = E$. It follows from the Baire category theorem that there exists some n_0 such that $\text{Int}(F_{n_0}) \neq \emptyset$. Thus $E = F_{n_0}$; absurd.

Example 1.10

Let $E = C[0, 1]$ be the Banach space of continuous real-valued functions on $[0, 1]$ provided with the sup norm. Then there exists a discontinuous linear functional on E . Indeed, let $(e_i)_{i \in I}$ be an algebraic basis of E such that $\|e_i\| = 1 \forall i \in I$. Since E is infinite-dimensional one may assume that $\mathbb{N} \subset I$. There exists a (unique) linear functional on E such that $f(e_i) = i$ if $i \in \mathbb{N}$ and $f(e_i) = 0$ if $i \in I \setminus \mathbb{N}$. This functional is clearly discontinuous.

Fact 1.11. Let E be a normed vector space. The following are equivalent:

- E is finite-dimensional.
- The closed unit ball of E is compact.
- Every linear functional on E is continuous.

§2 HW2

§2.1 Problem 1

Theorem 2.1

1.3 Let $E = \{u \in C([0, 1]; \mathbb{R}); u(0) = 0\}$ with its usual norm

$$\|u\| = \max_{t \in [0, 1]} |u(t)|.$$

Consider the linear functional

$$f : u \in E \mapsto f(u) = \int_0^1 u(t) dt.$$

1. Show that $f \in E^*$ and compute $\|f\|_{E^*}$.
2. Can one find some $u \in E$ such that $\|u\| = 1$ and $f(u) = \|f\|_{E^*}$?

Proof. 1. We first prove that $f \in E^*$.

Step 1: **Linearity.** For any $u, v \in E$ and $\alpha, \beta \in \mathbb{R}$, we have

$$f(\alpha u + \beta v) = \int_0^1 (\alpha u(t) + \beta v(t)) dt = \alpha \int_0^1 u(t) dt + \beta \int_0^1 v(t) dt = \alpha f(u) + \beta f(v).$$

Step 2: **Boundedness.** For any $u \in E$, we have

$$|f(u)| = \left| \int_0^1 u(t) dt \right| \leq \int_0^1 |u(t)| dt \leq \|u\| \int_0^1 dt = \|u\|.$$

Thus, we obtain

$$\|f\|_{E^*} = \sup_{\|u\| \leq 1} |f(u)| \leq 1.$$

Or we directly prove f is continuous. For any $u, v \in E$, we have

$$\begin{aligned} |f(u) - f(v)| &= \left| \int_0^1 (u(t) - v(t)) dt \right| \\ &\leq \int_0^1 |u(t) - v(t)| dt \\ &\leq \max_{t \in [0,1]} |u(t) - v(t)| \cdot \int_0^1 dt = \|u - v\|. \end{aligned}$$

So f satisfies the **Lipschitz condition** with constant 1. Then f is continuous.

We now compute $\|f\|_{E^*}$.

Use the definition of norm and **Intermediate Value Theorem**, we have:

$$\begin{aligned} \|f\|_{E^*} &= \sup_{u \in E} \frac{\langle f, u \rangle}{\|u\|} = \sup_{u \in E} \frac{\int_0^1 u(t) dt}{\max_{t \in [0,1]} |u(t)|} \\ &= \sup_{u \in E} \frac{u(\xi)}{\max_{t \in [0,1]} u(t)} (\xi \in (0, 1)) \leq 1 \end{aligned}$$

Now we prove another inequality to get the exact value of $\|f\|_{E^*}$.

Let $u(t) = t^\alpha$ for $\alpha > 0$. We compute $f(t^\alpha)$:

$$f(t^\alpha) = \int_0^1 t^\alpha dt = \frac{1}{\alpha + 1}.$$

Let $\alpha > 0$, $\forall \alpha > 0$ we have

$$\|f\|_{E^*} \geq \frac{f(t^\alpha)}{\|t^\alpha\|} = \frac{\frac{1}{\alpha+1}}{\max_{t \in [0,1]} |t^\alpha|} = \frac{1}{\alpha + 1}.$$

Letting $\alpha \rightarrow 0^+$, we obtain

$$\|f\|_{E^*} \geq 1.$$

Combining the two inequalities, we conclude that

$$\|f\|_{E^*} = 1.$$

2. Use the result in 1., we have $\|f\|_{E^*} = 1$. Suppose there exists some $u \in E$ such that $\|u\| = 1$ and $f(u) = \|f\|_{E^*} = 1$. Then we have $f(u) = \int_0^1 1 dt = 1$. Then we have

$$\int_0^1 (1 - u(t)) dt = 0.$$

Since $1 - u(t) \geq 0$ for all t in $[0, 1]$, we have $1 - u(t) = 0$ for all $t \in [0, 1]$. This means that $u(t) = 1$ for all $t \in [0, 1]$. However, this contradicts the condition that $u(0) = 0$. Thus, we conclude that there does not exist such a $u \in E$.

Remark 2.2. Solution on book P373. and Comments

1. Note that $f(t^\alpha) = 1/(\alpha + 1), \forall \alpha > 0$.

The key is how to find such $u = t^\alpha$!

$\Rightarrow$ Observe the graph of possible functions in E .

2. If there exists such a u we would have $\int_0^1 (1-u)dt = 0$ and thus $u \equiv 1$, absurd.

Why we cannot use the **corollary of Hahn-Banach Theorem** to get such u (E^* is a Banach space)?

AI says $\Rightarrow$ Because E is not a **reflexive space**, i.e., $E^{**} \neq E$.

□

§2.2 Problem 2**Theorem 2.3**

1.9 Hahn-Banach in finite-dimensional spaces Let E be a finite-dimensional normed space. Let $C \subset E$ be a nonempty convex set such that $0 \notin C$. We claim that there always exists some hyperplane that separates C and $\{0\}$.

[Note that every hyperplane is closed (why?). The main point in this exercise is that no additional assumption on C is required.]

1. Let $(x_n)_{n \geq 1}$ be a countable subset of C that is dense in C (why does it exist?). For every n let

$$C_n = \text{conv}\{x_1, x_2, \dots, x_n\} = \left\{ x = \sum_{i=1}^n t_i x_i; t_i \geq 0 \forall i : \text{and } \sum_{i=1}^n t_i = 1 \right\}.$$

Check that C_n is compact and that $\bigcup_{n=1}^\infty C_n$ is dense in C .

2. Prove that there is some $f_n \in E^*$ such that

$$\|f_n\| = 1 \quad \text{and} \quad \langle f_n, x \rangle \geq 0 \quad \forall x \in C_n.$$

3. Deduce that there is some $f \in E^*$ such that

$$\|f\| = 1 \quad \text{and} \quad \langle f, x \rangle \geq 0 \quad \forall x \in C.$$

Conclude.

4. Let $A, B \subset E$ be nonempty disjoint convex sets. Prove that there exists some hyperplane H that separates A and B .

Proof. 1. Since E is a finite-dimensional normed space, it is separable. Thus, there exists a countable subset $(x_n)_{n \geq 1}$ of C that is dense in C . For each $n \geq 1$, we define

$$C_n = \text{conv}\{x_1, x_2, \dots, x_n\} = \left\{ x = \sum_{i=1}^n t_i x_i; t_i \geq 0 \forall i, \text{ and } \sum_{i=1}^n t_i = 1 \right\}.$$

We first check that C_n is **compact**. Let

$$T = \left\{ t = (t_1, t_2, \dots, t_n) \in \mathbb{R}^n \mid t_i \geq 0 (\forall i), \quad \sum_{i=1}^n t_i = 1 \right\}.$$

Then C_n can be identified with the image of the compact set T under the map

$$p = P_{\{x_1, \dots, x_n\}} : T \rightarrow C_n$$

$$t \mapsto x = \sum_{i=1}^n t_i x_i.$$

We now show that p is continuous. For any $t, t' \in T$, we have

$$\begin{aligned} \|p(t) - p(t')\| &= \left\| \sum_{i=1}^n (t_i - t'_i) x_i \right\| \\ &\leq \sum_{i=1}^n |t_i - t'_i| \|x_i\| \\ &\leq \left(\max_{1 \leq i \leq n} \|x_i\| \right) \sum_{i=1}^n |t_i - t'_i| \\ &\leq \left(\max_{1 \leq i \leq n} \|x_i\| \right) \sqrt{n} \|t - t'\| \end{aligned}$$

where the last inequality follows from the Cauchy-Schwarz inequality. Thus, p is continuous.

Since T is compact and p is continuous, the image $C_n = p(T)$ is also compact.

2. We next prove that there exists some $f_n \in E^*$ such that

$$\|f_n\| = 1 \quad \text{and} \quad \langle f_n, x \rangle \geq 0 \quad \forall x \in C_n.$$

Since $0 \notin C_n$ and C_n is compact, there exists some $x_0 \in C_n$ such that

$$\|x_0\| = \min_{x \in C_n} \|x\| > 0.$$

By the **Hahn-Banach theorem**, there exists some $f_n \in E^*$ such that

$$\|f_n\| = 1 \quad \text{and} \quad \langle f_n, x_0 \rangle = \|x_0\| > 0.$$

We now show that $\langle f_n, x \rangle \geq 0$ for all $x \in C_n$. Suppose there exists some $x_1 \in C_n$ such that $\langle f_n, x_1 \rangle < 0$. Then for any $t \in (0, 1)$, we have

$$\langle f_n, tx_0 + (1-t)x_1 \rangle = t\langle f_n, x_0 \rangle + (1-t)\langle f_n, x_1 \rangle.$$

Since $\langle f_n, x_0 \rangle > 0$ and $\langle f_n, x_1 \rangle < 0$, we can choose t sufficiently small such that

$$\langle f_n, tx_0 + (1-t)x_1 \rangle < 0.$$

Note that $tx_0 + (1-t)x_1 \in C_n$ since C_n is convex. This contradicts the fact that $\|x_0\| = \min_{x \in C_n} \|x\|$. Thus, we conclude that $\langle f_n, x \rangle \geq 0$ for all $x \in C_n$. 3. We now deduce that there exists some $f \in E^*$ such that

$$\|f\| = 1 \quad \text{and} \quad \langle f, x \rangle \geq 0 \quad \forall x \in C.$$

Since E is a finite-dimensional normed space, E^* is also a finite-dimensional normed space. Thus, the unit ball in E^* is compact. Therefore, the sequence $(f_n)_{n \geq 1}$ has a

convergent subsequence $(f_{n_k})_{k \geq 1}$ that converges to some $f \in E^*$. Since $\|f_{n_k}\| = 1$ for all $k \geq 1$, we have $\|f\| = 1$. We now show that $\langle f, x \rangle \geq 0$ for all $x \in C$. For any $x \in C$, since $\bigcup_{n=1}^{\infty} C_n$ is dense in C , there exists a sequence $(x_m)_{m \geq 1} \subset \bigcup_{n=1}^{\infty} C_n$ that converges to x . For each $m \geq 1$, there exists some $N_m \geq 1$ such that $x_m \in C_{N_m}$. Since $\langle f_{n_k}, y \rangle \geq 0$ for all $y \in C_{n_k}$ and all $k \geq 1$, we have

$$\langle f_{n_k}, x_m \rangle \geq 0 \quad \forall k \geq 1.$$

Letting $k \rightarrow \infty$, we get

$$\langle f, x_m \rangle \geq 0 \quad \forall m \geq 1.$$

Letting $m \rightarrow \infty$, we obtain

$$\langle f, x \rangle \geq 0.$$

Since x was arbitrary, we conclude that $\langle f, x \rangle \geq 0$ for all $x \in C$. Thus, we have found some $f \in E^*$ such that

$$\|f\| = 1 \quad \text{and} \quad \langle f, x \rangle \geq 0 \quad \forall x \in C.$$

This means that the hyperplane

$$H = \{x \in E; \langle f, x \rangle = 0\}$$

separates C and $\{0\}$. 4. We finally prove that there exists some hyperplane H that separates A and B .

Since A and B are nonempty disjoint convex sets, we have

$$0 \notin A - B = \{a - b; a \in A, b \in B\}.$$

Note that $A - B$ is also a convex set. By the previous result, there exists some hyperplane H that separates $A - B$ and $\{0\}$. This means that there exists some $f \in E^*$ and $\alpha \in \mathbb{R}$ such that

$$H = \{x \in E; \langle f, x \rangle = \alpha\}$$

$$\langle f, a - b \rangle \geq \alpha \quad \forall a \in A, b \in B.$$

We now show that H separates A and B . For any $a \in A$ and $b \in B$, we have

$$\langle f, a \rangle = \langle f, (a - b) + b \rangle = \langle f, a - b \rangle + \langle f, b \rangle \geq \alpha + \langle f, b \rangle.$$

This implies that

$$\langle f, a \rangle \geq \alpha + \inf_{b \in B} \langle f, b \rangle \quad \forall a \in A.$$

Similarly, we have

$$\langle f, b \rangle \leq \sup_{a \in A} \langle f, a \rangle - \alpha \quad \forall b \in B.$$

Letting $b \rightarrow \infty$, we obtain

$$\langle f, b \rangle \leq \sup_{a \in A} \langle f, a \rangle - \alpha.$$

Since A and B are disjoint, we have

$$\inf_{b \in B} \langle f, b \rangle > \sup_{a \in A} \langle f, a \rangle - \alpha.$$

Thus, we conclude that H separates A and B .

□

§2.3 Problem 3

Theorem 2.4

1.14 Let $E = \ell^1$ (see Section 11.3) and consider the two sets

$$X = \{x = (x_n)_{n \geq 1} \in E; x_{2n} = 0 \ \forall n \geq 1\}$$

and

$$Y = \left\{ y = (y_n)_{n \geq 1} \in E; y_{2n} = \frac{1}{2^n} y_{2n-1} \ \forall n \geq 1 \right\}.$$

1. Check that X and Y are closed linear spaces and that $\overline{X + Y} = E$.
2. Let $c \in E$ be defined by

$$\begin{cases} c_{2n-1} = 0 & \forall n \geq 1, \\ c_{2n} = \frac{1}{2^n} & \forall n \geq 1. \end{cases}$$

Check that $c \notin X + Y$.

3. Set $Z = X - c$ and check that $Y \cap Z = \emptyset$. Does there exist a closed hyperplane in E that separates Y and Z ? Compare with Theorem 1.7 and Exercise 1.9.
4. Same questions in $E = \ell^p$, $1 < p < \infty$, and in $E = c_0$.

Proof. 1.

$$\ell^1 = \{x = (x_1, x_2, \dots, x_k, \dots) \mid \|x\|_1 = \sum_{k=1}^{\infty} |x_k| < \infty\}$$

We first check that X and Y are closed linear spaces.

For any $x, x' \in X$ and $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha x + \beta x')_{2n} = \alpha x_{2n} + \beta x'_{2n} = 0 + 0 = 0,$$

so $\alpha x + \beta x' \in X$. Thus, X is a linear space.

For any sequence $(x^m)_{m \geq 1} \subset X$ that converges to some $x \in E$, we have

$$x_{2n}^m = 0 \quad \forall m, n \geq 1.$$

Since the convergence in ℓ^1 implies pointwise convergence, we have

$$x_{2n} = \lim_{m \rightarrow \infty} x_{2n}^m = 0 \quad \forall n \geq 1.$$

Thus, $x \in X$, so X is closed.

Now we check the Y . For any $y, y' \in Y$ and $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha y + \beta y')_{2n} = \alpha y_{2n} + \beta y'_{2n} = \alpha \frac{1}{2^n} y_{2n-1} + \beta \frac{1}{2^n} y'_{2n-1} = \frac{1}{2^n} (\alpha y_{2n-1} + \beta y'_{2n-1}),$$

so $\alpha y + \beta y' \in Y$. Thus, Y is a linear space.

For any sequence $(y^m)_{m \geq 1} \subset Y$ that converges to some $y \in E$, we have

$$y_{2n}^m = \frac{1}{2^n} y_{2n-1}^m \quad \forall m, n \geq 1.$$

Since the convergence in ℓ^1 implies pointwise convergence, we have

$$y_{2n} = \lim_{m \rightarrow \infty} y_{2n}^m = \frac{1}{2^n} y_{2n-1} \quad \forall n \geq 1.$$

Thus, $y \in Y$, so Y is closed.

We next show that $X + Y$ is dense in E .

Let $z = (z_n)_{n \geq 1} \in E$ and $\epsilon > 0$. We can choose $N \in \mathbb{N}$ such that

$$\sum_{n=N+1}^{\infty} |z_n| < \frac{\epsilon}{2}.$$

We then define $x = (x_n)_{n \geq 1} \in X$ and $y = (y_n)_{n \geq 1} \in Y$ by

$$x_{2n} = 0, \quad x_{2n-1} = z_{2n-1} \quad \forall n = 1, 2, \dots, N,$$

$$x_n = 0 \quad \forall n > 2N,$$

$$y_{2n} = z_{2n} \quad \forall n = 1, 2, \dots, N,$$

$$y_{2n-1} = 2^n z_{2n} \quad \forall n = 1, 2, \dots, N,$$

$$y_n = 0 \quad \forall n > 2N.$$

Then we have

$$z = x + y \in X + Y.$$

Since z was arbitrary, we conclude that $X + Y$ is dense in E .

2. We now check that $c \notin X + Y$.

Suppose there exist $x \in X$ and $y \in Y$ such that $c = x + y$. Then we have

$$c_{2n-1} = x_{2n-1} + y_{2n-1} = 0 + 0 = 0 \quad \forall n \geq 1,$$

$$c_{2n} = x_{2n} + y_{2n} = 0 + \frac{1}{2^n} = \frac{1}{2^n} \quad \forall n \geq 1.$$

This means that

$$c = (0, \frac{1}{2^1}, 0, \frac{1}{2^2}, 0, \frac{1}{2^3}, \dots) \in X + Y.$$

However, this contradicts the definition of X and Y . Thus, we conclude that $c \notin X + Y$.

3. We set $Z = X - c$ and check that $Y \cap Z = \emptyset$.

Suppose there exists some $z \in Y \cap Z$. Then we have

$$z \in Y \implies z_{2n} = \frac{1}{2^n} z_{2n-1} \quad \forall n \geq 1,$$

$$z \in Z \implies z = x - c \text{ for some } x \in X \implies z_{2n} = x_{2n} - c_{2n} = 0 - \frac{1}{2^n} = -\frac{1}{2^n} \quad \forall n \geq 1.$$

Combining these two results, we get

$$-\frac{1}{2^n} = \frac{1}{2^n} z_{2n-1} \implies z_{2n-1} = -1 \quad \forall n \geq 1.$$

This means that

$$z = (-1, \frac{1}{2^1}, -1, \frac{1}{2^2}, -1, \frac{1}{2^3}, \dots) \in Y \cap Z.$$

Thus, we conclude that $Y \cap Z = \emptyset$. We next check whether there exists a closed hyperplane in E that separates Y and Z .

Suppose there exists a closed hyperplane H in E that separates Y and Z . Then there exists some $f \in E^*$ and $\alpha \in \mathbb{R}$ such that

$$H = \{x \in E; f(x) = \alpha\},$$

$$f(y) \leq \alpha \quad \forall y \in Y,$$

$$f(z) \geq \alpha \quad \forall z \in Z.$$

Since Y and Z are linear spaces, we have

$$f(y) = 0 \quad \forall y \in Y,$$

$$f(z) = 0 \quad \forall z \in Z.$$

Thus, we have $\alpha = 0$ and

$$f(y) = 0 \quad \forall y \in Y,$$

$$f(z) = 0 \quad \forall z \in Z.$$

This means that f vanishes on both Y and Z . Since $\overline{X + Y} = E$, we have

$$f(x) = 0 \quad \forall x \in E.$$

This implies that f is the zero functional, which contradicts the fact that H is a hyperplane. Thus, we conclude that there does not exist a closed hyperplane in E that separates Y and Z . 4. We now consider the same questions in $E = \ell^p$, $1 < p < \infty$, and in $E = c_0$.

For $E = \ell^p$, $1 < p < \infty$, we can similarly define the sets

$$X = \{x = (x_n)_{n \geq 1} \in E; x_{2n} = 0 \quad \forall n \geq 1\},$$

$$Y = \left\{ y = (y_n)_{n \geq 1} \in E; y_{2n} = \frac{1}{2^n} y_{2n-1} \quad \forall n \geq 1 \right\},$$

and the element $c = (0, \frac{1}{2^1}, 0, \frac{1}{2^2}, 0, \frac{1}{2^3}, \dots) \in E$. We can then check that X and Y are closed linear spaces and that $\overline{X + Y} = E$. We can also check that $c \notin X + Y$ and that $Y \cap Z = \emptyset$ where $Z = X - c$. However, unlike the case of ℓ^1 , there does exist a closed hyperplane in E that separates Y and Z . This is because ℓ^p is a reflexive Banach space for $1 < p < \infty$, and thus the Hahn-Banach theorem guarantees the existence of such a hyperplane. For $E = c_0$, we can similarly define the sets

$$X = \{x = (x_n)_{n \geq 1} \in E; x_{2n} = 0 \quad \forall n \geq 1\},$$

$$Y = \left\{ y = (y_n)_{n \geq 1} \in E; y_{2n} = \frac{1}{2^n} y_{2n-1} \quad \forall n \geq 1 \right\},$$

and the element $c = (0, \frac{1}{2^1}, 0, \frac{1}{2^2}, 0, \frac{1}{2^3}, \dots) \in E$. We can then check that X and Y are closed linear spaces and that $\overline{X + Y} = E$. We can also check that $c \notin X + Y$ and that $Y \cap Z = \emptyset$ where $Z = X - c$. However, unlike the case of ℓ^1 , there does exist a closed hyperplane in E that separates Y and Z . This is because c_0 is a Banach space and thus the Hahn-Banach theorem guarantees the existence of such a hyperplane. □

§3 HW3

§3.1 Problem 1

Theorem 3.1

2.3 Let E and F be two Banach spaces and let (T_n) be a sequence in $\mathcal{L}(E, F)$. Assume that for every $x \in E$, $T_n x$ converges as $n \rightarrow \infty$ to a limit denoted by Tx . Show that if $x_n \rightarrow x$ in E , then $T_n x_n \rightarrow Tx$ in F .

Proof. By the **Corollary 2.3 of the Uniform Boundedness Principle**, we have $\sup_n \|T_n\| = c < \infty$.

At the same time, we notice that:

$$\begin{aligned} \|T_n x_n - Tx\| &\leq \|T_n x_n - T_n x\| + \|T_n x - Tx\| \\ &\leq \sup_n \|T_n\| \cdot \|x_n - x\| + \|T_n x - Tx\| \rightarrow 0 \\ &\leq c \|x_n - x\| + \|T_n x - Tx\| \rightarrow 0 \quad (n \rightarrow \infty) \end{aligned}$$

□

§3.2 Problem 2**Theorem 3.2**

2.4 Let E and F be two Banach spaces and let $a : E \times F \rightarrow \mathbb{R}$ be a bilinear form satisfying:

- (i) for each fixed $x \in E$, the map $y \mapsto a(x, y)$ is continuous;
- (ii) for each fixed $y \in F$, the map $x \mapsto a(x, y)$ is continuous.

Prove that there exists a constant $C \geq 0$ such that

$$|a(x, y)| \leq C \|x\| \|y\| \quad \forall x \in E, \quad \forall y \in F.$$

[Hint: Introduce a linear operator $T : E \rightarrow F^*$ and prove that T is bounded with the help of Corollary 2.5.]

Proof. Define the linear operator $T : E \rightarrow F^*$ by $(Tx)(y) = a(x, y)$ for all $y \in F$. Now we show that T is well-defined. For fixed x, y , we want to prove that $(Tx)y = (T'x)y \Rightarrow T = T'$. For any $z \in F$, we have:

$$T(x + y)(z) = a(x + y, z) = a(x, z) + a(y, z) = (Tx)(z) + (Ty)(z) = (Tx + Ty)(z).$$

$$(T(\alpha x))(y) = a(\alpha x, y) = \alpha a(x, y) = \alpha (Tx)(y).$$

So $(Tx)(\cdot)$ is linear.

Now we see that T is well-defined. $(Tx - T'x)y = 0$ for all $y \in F$. Thus, $Tx = T'x$.

For each fixed $x \in E$, condition (i) implies that Tx is continuous, so $Tx \in F^*$. For each fixed $y \in F$, condition (ii) implies that the map $x \mapsto (Tx)(y)$ is continuous. Thus, for each $y \in F$, the set $(Tx)(y) : x \in E, |x| = 1$ is bounded. By the Uniform Boundedness Principle (Corollary 2.5), there exists a constant $C \geq 0$ such that $|Tx| \leq C|x|$ for all $x \in E$. Then for any $x \in E, y \in F$:

$$|a(x, y)| = |(Tx)(y)| \leq \|Tx\| \cdot \|y\| \leq C \|x\| \cdot \|y\|.$$

□

Remark 3.3. 2.4 By (i) there exists a linear operator $T : E \rightarrow F^*$ such that $a(x, y) = \langle Tx, y \rangle_{F^*, F} \forall x, y$. The aim is to show that T is a bounded operator, i.e., $T(B_E)$ is bounded in F^* . In view of Corollary 2.5 it suffices to fix $y \in F$ and to check that $\langle T(B_E), y \rangle$ is bounded. This follows from (ii).

§3.3 Problem 3

Theorem 3.4

2.5 Let E be a Banach space and let ε_n be a sequence of positive numbers such that $\lim \varepsilon_n = 0$. Further, let (f_n) be a sequence in E^* satisfying the property

$$\begin{cases} \exists r > 0, \forall x \in E \text{ with } \|x\| < r, \exists C(x) \in \mathbb{R} \text{ such that} \\ \langle f_n, x \rangle \leq \varepsilon_n \|f_n\| + C(x), \forall n. \end{cases}$$

Prove that (f_n) is bounded. [Hint: Introduce $g_n = f_n/(1 + \varepsilon_n \|f_n\|)$.]

Proof. Assume, for contradiction, that (f_n) is unbounded. Define $g_n = f_n/(1 + \varepsilon_n \|f_n\|)$. Then $|g_n| = \|f_n\|/(1 + \varepsilon_n \|f_n\|)$. Given $r > 0$, for any $x \in E$ with $\|x\| < r$, there exists $C(x) \in \mathbb{R}$ such that:

$$\langle f_n, x \rangle \leq \varepsilon_n \|f_n\| + C(x).$$

Replacing x by $-x$, we also have:

$$-\langle f_n, x \rangle \leq \varepsilon_n \|f_n\| + C(-x) \implies \langle f_n, x \rangle \geq -\varepsilon_n \|f_n\| - C(-x).$$

Dividing by $1 + \varepsilon_n \|f_n\| > 0$:

$$\frac{-\varepsilon_n \|f_n\| - C(-x)}{1 + \varepsilon_n \|f_n\|} \leq \langle g_n, x \rangle \leq \frac{\varepsilon_n \|f_n\| + C(x)}{1 + \varepsilon_n \|f_n\|}.$$

Note that $\frac{\varepsilon_n \|f_n\|}{1 + \varepsilon_n \|f_n\|} \leq 1$. Hence,

$$-1 - C(-x) \leq \langle g_n, x \rangle \leq 1 + C(x).$$

So for each x with $\|x\| < r$, $\sup_n |\langle g_n, x \rangle| < \infty$. For any $x \in E$, $x \neq 0$, let $y = (r/2)(x/\|x\|)$, then $\|y\| < r$, and $\langle g_n, x \rangle = (2\|x\|/r)\langle g_n, y \rangle$, so $\sup_n |\langle g_n, x \rangle| < \infty$. By the Uniform Boundedness Principle, $\sup_n \|g_n\| < \infty$. But if (f_n) is unbounded, then there exists a subsequence with $\|f_n\| \rightarrow \infty$. For such n , $|g_n| = \|f_n\|/(1 + \varepsilon_n \|f_n\|)$. Since $\varepsilon_n \rightarrow 0$, for large n , $1 + \varepsilon_n \|f_n\| \sim \varepsilon_n \|f_n\|$, so $|g_n| \sim 1/\varepsilon_n \rightarrow \infty$, contradiction. Hence, (f_n) is bounded. $\square$

§3.4 Problem 4

Theorem 3.5

[2.17] Let $E = C([0, 1])$ with its usual norm. Consider the operator $A : D(A) \subset E \rightarrow E$ defined by

$$D(A) = C^1([0, 1]) \quad \text{and} \quad Au = u' = \frac{du}{dt}.$$

1. Check that $\overline{D(A)} = E$.
2. Is A closed?
3. Consider the operator $B : D(B) \subset E \rightarrow E$ defined by

$$D(B) = C^2([0, 1]) \quad \text{and} \quad Bu = u' = \frac{du}{dt}.$$

Is B closed?

Proof. 1. $\overline{D(A)} = E$ because polynomials are dense in $C([0, 1])$ (by Weierstrass approximation) and polynomials are in $C^1([0, 1])$.

2. A is closed. Let $(u_n) \subset D(A)$ such that $u_n \rightarrow u$ uniformly and $Au_n = u'_n \rightarrow v$ uniformly. Then by standard calculus, u is differentiable and $u' = v$. Since v is continuous, $u \in C^1([0, 1]) = D(A)$ and $Au = v$. So A is closed.

3. B is not closed. Consider $u(x) = \int_0^x |s - 1/2| ds$. Then $u \in C^1([0, 1])$ but $u \notin C^2([0, 1])$ because $u'(x) = |x - 1/2|$ is not differentiable at $x = 1/2$. By Weierstrass approximation, there exists a sequence of polynomials $(u_n) \subset C^2([0, 1])$ such that $u_n \rightarrow u$ uniformly and $u'_n \rightarrow u'$ uniformly. Then $u_n \in D(B)$, $u_n \rightarrow u$, and $Bu_n = u'_n \rightarrow u'$ uniformly, but $u \notin D(B)$. Hence, B is not closed. $\square$

§4 HW4**§4.1 Problem 1****Theorem 4.1**

[3.1] Let E be a Banach space and let $A \subset E$ be a subset that is compact in the weak topology $\sigma(E, E^*)$. Prove that A is bounded.

Proof. Since A is compact in the weak topology, and $\forall f \in E^*$, **the linear functional f is continuous in the weak topology**, we have that $f(A)$ is compact in $\mathbb{R}$. **(Continuous maps preserve compactness.)** Thus, $f(A)$ is bounded in $\mathbb{R}$, i.e. $\forall f \in E^*$, we have:

$$\sup_{a \in A} |f(a)| < \infty.$$

Consider the canonical embedding $J : E \rightarrow E^{**}$ defined by $(Jx)(f) = f(x)$ for all $f \in E^*$. For each fixed $a \in A$, we have:

$$\sup_{f \in E^*, \|f\| \leq 1} |(Ja)(f)| = \sup_{f \in E^*, \|f\| \leq 1} |f(a)| = \|a\|.$$

Thus, for each fixed $a \in A$, the set $\{(Ja)(f) : f \in E^*, \|f\| \leq 1\}$ is bounded. By the **Uniform Boundedness Principle**, there exists a constant $C \geq 0$ such that $\|Ja\| \leq C$ for all $a \in A$. Since J is an isometry, we have $\|a\| = \|Ja\| \leq C$ for all $a \in A$. Therefore, A is bounded. $\square$

Remark 4.2. 3.1 Apply Corollary 2.4.

§4.2 Problem 2

Theorem 4.3

3.2 Let E be a Banach space and let (x_n) be a sequence such that $x_n \rightharpoonup x$ in the weak topology $\sigma(E, E^*)$. Set

$$\sigma_n = \frac{1}{n}(x_1 + x_2 + \cdots + x_n).$$

Prove that $\sigma_n \rightharpoonup x$ in the weak topology $\sigma(E, E^*)$.

Proof. By **Proposition 3.5 in P58**: $[x_n \rightharpoonup x] \Leftrightarrow [\forall f \in E^*, f(x_n) \rightarrow f(x)]$, we have that for each $f \in E^*$, $f(x_n) \rightarrow f(x)$ as $n \rightarrow \infty$. Thus, $\forall f \in E^*$, we have:

$$f(\sigma_n) = f\left(\frac{1}{n}(x_1 + x_2 + \cdots + x_n)\right) = \frac{1}{n}(f(x_1) + f(x_2) + \cdots + f(x_n)) \rightarrow f(x) \quad (n \rightarrow \infty).$$

Remark 4.4. To proof the last step, we use the **Cesàro mean theorem**: If a sequence (a_n) converges to a limit a , then the sequence of its arithmetic means also converges to the same limit, i.e.

$$\frac{1}{n}(a_1 + a_2 + \cdots + a_n) \rightarrow a \quad (n \rightarrow \infty).$$

3.2 Note that $\langle f, \sigma_n \rangle = \frac{1}{n} \sum_{i=1}^n \langle f, x_i \rangle \quad \forall f \in E^*$. Since $\langle f, x_n \rangle \rightarrow \langle f, x \rangle$, it follows that $\langle f, \sigma_n \rangle \rightarrow \langle f, x \rangle$.

□

§4.3 Problem 3

Theorem 4.5

3.5 Let E be a Banach space and let $K \subset E$ be a subset of E that is compact in the strong topology. Let (x_n) be a sequence in K such that $x_n \rightharpoonup x$ weakly $\sigma(E, E^*)$. Prove that $x_n \rightarrow x$ strongly. [**Hint: Argue by contradiction.**]

Proof. We argue by contradiction. Suppose that $x_n \not\rightarrow x$ strongly. Then $\exists \varepsilon_0 > 0$ such that for infinitely many n , we have $\|x_n - x\| \geq \varepsilon_0$. Thus, we can extract a subsequence (x_{n_k}) from (x_n) such that $\|x_{n_k} - x\| \geq \varepsilon_0$ for all k .

Since K is compact in the strong topology, the sequence (x_{n_k}) has a convergent subsequence $(x_{n_{k_j}})$ that converges strongly to some point $y \in K$. That is, $x_{n_{k_j}} \rightarrow y$ strongly as $j \rightarrow \infty$.

On the other hand, since $x_n \rightharpoonup x$ weakly, we have $x_{n_{k_j}} \rightharpoonup x$ weakly as well. By the **uniqueness of weak limits in Banach spaces**, we must have $y = x$.

However, this contradicts our earlier assumption that $\|x_{n_k} - x\| \geq \varepsilon$ for all k , since if $x_{n_{k_j}} \rightarrow y = x$ strongly, then $\|x_{n_{k_j}} - x\| \rightarrow 0$.

Therefore, our initial assumption must be false, and we conclude that $x_n \rightarrow x$ strongly.

□

§4.4 Problem 4

Theorem 4.6

3.6 Let X be a topological space and let E be a Banach space. Let $u, v : X \rightarrow E$ be two continuous maps from X with values in E equipped with the weak topology $\sigma(E, E^*)$.

1. Prove that the map $x \mapsto u(x) + v(x)$ is continuous from X into E equipped with $\sigma(E, E^*)$.
2. Let $a : X \rightarrow \mathbb{R}$ be a continuous function. Prove that the map $x \mapsto a(x)u(x)$ is continuous from X into E equipped with $\sigma(E, E^*)$.

Proof. 1. Let $f \in E^*$. We need to show that $f(u(x) + v(x))$ is continuous as a function of $x \in X$. Since u and v are continuous, we have $u(x) \in E$ and $v(x) \in E$ for all $x \in X$. Thus, we can write:

$$f(u(x) + v(x)) = f(u(x)) + f(v(x)).$$

Since f is a continuous linear functional on E , and both u and v are continuous maps from X to E equipped with the weak topology, it follows that $f(u(x))$ and $f(v(x))$ are continuous functions of x . Therefore, their sum $f(u(x) + v(x))$ is also continuous. Since this holds for all $f \in E^*$, we conclude that the map $x \mapsto u(x) + v(x)$ is continuous from X into E equipped with the weak topology $\sigma(E, E^*)$.

2. Let $a : X \rightarrow \mathbb{R}$ be a continuous function. We need to show that $f(a(x)u(x))$ is continuous as a function of $x \in X$. We can write:

$$f(a(x)u(x)) = a(x)f(u(x)).$$

Since $a(x)$ is continuous and $f(u(x))$ is continuous (as shown in the first part), their product $a(x)f(u(x))$ is also continuous. Since this holds for all $f \in E^*$, we conclude that the map $x \mapsto a(x)u(x)$ is continuous from X into E equipped with the weak topology $\sigma(E, E^*)$. $\square$

§5 HW5

§5.1 Problem 1

Theorem 5.1

3.3 Let E be a Banach space. Let $A \subset E$ be a convex subset. Prove that the closure $\overline{A}$ in the strong topology and that in the weak topology $\sigma(E, E^*)$ are the same.

Proof. Since the strong topology is finer than the weak topology, we have $\overline{A}^{\text{strong}} \subset \overline{A}^{\sigma(E, E^*)}$.

To prove the reverse inclusion, let $x \in \overline{A}^{\sigma(E, E^*)}$. We want to show that $x \in \overline{A}^{\text{strong}}$. Suppose, by contradiction, that $x \notin \overline{A}^{\text{strong}}$. Since $\overline{A}^{\text{strong}}$ is closed and convex, by the **Hahn-Banach separation theorem**, there exists a continuous linear functional $f \in E^*$ and a real number α such that:

$$f(x) > \alpha \geq f(a) \quad \forall a \in \overline{A}^{\text{strong}}.$$

However, since $x \in \overline{A}^{\sigma(E, E^*)}$, there exists a net $(a_i) \subset A$ such that $a_i \rightarrow x$ in the weak topology. This implies that:

$$f(a_i) \rightarrow f(x)$$

But this contradicts the fact that $f(a_i) \leq \alpha < f(x)$ for all i . Therefore, our assumption that $x \notin \overline{A}^{\text{strong}}$ must be false, and we conclude that $x \in \overline{A}^{\text{strong}}$. Hence, $\overline{A}^{\sigma(E, E^*)} \subset \overline{A}^{\text{strong}}$. Thus, we have $\overline{A}^{\text{strong}} = \overline{A}^{\sigma(E, E^*)}$. □

Remark 5.2. Note that $\langle f, \sigma_n \rangle = \frac{1}{n} \sum_{i=1}^n \langle f, x_i \rangle \forall f \in E^*$. Since $\langle f, x_n \rangle \rightarrow \langle f, x \rangle$, it follows that $\langle f, \sigma_n \rangle \rightarrow \langle f, x \rangle$.

§5.2 Problem 2

Theorem 5.3

3.7 Let E be a Banach space and let $A \subset E$ be a subset that is closed in the weak topology $\sigma(E, E^*)$. Let $B \subset E$ be a subset that is compact in the weak topology $\sigma(E, E^*)$.

1. Prove that $A + B$ is closed in $\sigma(E, E^*)$.
2. Assume, in addition, that A and B are convex, nonempty, and disjoint. Prove that there exists a closed hyperplane strictly separating A and B .

Proof. 1. Let $(x_n) \subset A + B$ be a net that converges to some $x \in E$ in the weak topology $\sigma(E, E^*)$. Then for each n , we can write $x_n = a_n + b_n$ with $a_n \in A$ and $b_n \in B$. Since A is closed in the weak topology, it follows that $a_n \rightarrow a \in A$ for some $a \in A$.

On the other hand, since B is compact in the weak topology, we can extract a subnet (b_{n_k}) that converges weakly to some $b \in B$. Thus, we have:

$$x = a + b \in A + B.$$

Therefore, $A + B$ is closed in the weak topology $\sigma(E, E^*)$.

2. Assume that A and B are convex, nonempty, and disjoint. By the separating hyperplane theorem, there exists a continuous linear functional $f \in E^*$ such that:

$$f(a) < 0 \quad \forall a \in A, \quad f(b) > 0 \quad \forall b \in B.$$

This functional f defines a closed hyperplane given by the equation $f(x) = 0$, which strictly separates the sets A and B . □

Remark 5.4. 3.7. 1. Let $x \notin A + B$. We shall construct a neighborhood W of 0 for $\sigma(E, E^*)$ such that

$$(x + W) \cap (A + B) = \emptyset.$$

For every $y \in B$ there exists a convex neighborhood $V(y)$ of 0 such that

$$(x + V(y)) \cap (A + y) = \emptyset$$

(since $A + y$ is closed and $x \notin A + y$). Clearly

$$B \subset \bigcup_{y \in B} \left(y - \frac{1}{2}V(y) \right),$$

and since B is compact, there is some finite set I such that

$$B \subset \bigcup_{i \in I} \left(y_i - \frac{1}{2} V(y_i) \right) \text{ with } y_i \in B.$$

Set

$$W = \frac{1}{2} \bigcap_{i \in I} V(y_i).$$

We claim that $(x + W) \cap (A + B) = \emptyset$. Indeed, suppose by contradiction that there exists some $w \in W$ such that $x + w \in (A + B)$. Hence there is some $i \in I$ such that

$$x + w \in A + y_i - \frac{1}{2} V(y_i).$$

Since $V(y_i)$ is convex it follows that there exists some $w' \in V(y_i)$ such that $x + w' \in A + y_i$. Consequently $(x + V(y_i)) \cap (A + y_i) \neq \emptyset$; absurd.

Remark. If E^* is separable and A is bounded one may use sequences in order to prove that $A + B$ is closed, since the weak topology is metrizable on bounded sets (see Theorem 3.29). This makes the argument somewhat easier. Indeed, let $x_n = a_n + b_n$ be a sequence such that $x_n \rightharpoonup x$ weakly $\sigma(E, E^*)$ with $a_n \in A$ and $b_n \in B$. Since B is weakly compact (and metrizable), there is a subsequence such that $b_{n_k} \rightharpoonup b$ weakly $\sigma(E, E^*)$ with $b \in B$. Thus $a_{n_k} \rightharpoonup x - b$ weakly $\sigma(E, E^*)$. But A is weakly closed and therefore $x - b \in A$, i.e., $x \in A + B$. By question 1, $(A - B)$ is weakly closed and therefore it is strongly closed. Hence one may strictly separate $\{0\}$ and $(A - B)$.

§5.3 Problem 3

Theorem 5.5

[3.8] Let E be an infinite-dimensional Banach space. Our purpose is to show that E equipped with the weak topology is not metrizable. Suppose, by contradiction, that there is a metric $d(x, y)$ on E that induces on E the same topology as $\sigma(E, E^*)$.

1. For every integer $k \geq 1$ let V_k denote a neighborhood of 0 in the topology $\sigma(E, E^*)$, such that

$$V_k \subset \left\{ x \in E; d(x, 0) < \frac{1}{k} \right\}.$$

Prove that there exists a sequence (f_n) in E^* such that every $g \in E^*$ is a (finite) linear combination of the f_n 's. [Hint: Use Lemma 3.2.]

2. Deduce that E^* is finite-dimensional. [Hint: Use the Baire category theorem as in Exercise 1.5.]

3. Conclude.

4. Prove by a similar method that E^* equipped with the weak* topology $\sigma(E^*, E)$ is not metrizable.

Proof. 1. For each $k \geq 1$, there exist finitely many functionals $f_{k,1}, f_{k,2}, \dots, f_{k,n_k} \in E^*$ and $\epsilon_k > 0$ such that:

$$V_k \supset \{x \in E : |f_{k,i}(x)| < \epsilon_k, i = 1, 2, \dots, n_k\}.$$

Let (f_n) be the sequence obtained by listing all the functionals $f_{k,i}$ for $k \geq 1$ and $1 \leq i \leq n_k$. We claim that every $g \in E^*$ can be expressed as a finite linear combination

of the f_n 's. Suppose not. Then there exists $g \in E^*$ that is not in the span of the f_n 's. Since g is continuous, there exists a neighborhood U of 0 in the weak topology such that $|g(x)| < 1$ for all $x \in U$. By the construction of V_k , there exists k such that $V_k \subset U$. Thus, for all $x \in V_k$, we have $|g(x)| < 1$. However, since g is not in the span of the f_n 's, we can find $x \in V_k$ such that $|g(x)| \geq 1$, a contradiction. Hence, every $g \in E^*$ is a finite linear combination of the f_n 's.

2. Since every $g \in E^*$ is a finite linear combination of the f_n 's, it follows that E^* is spanned by the countable set $\{f_n\}$. By the Baire category theorem, if E^* were infinite-dimensional, it would not be possible for a countable set to span it. Therefore, E^* must be finite-dimensional.

3. Since E^* is finite-dimensional, by the Riesz representation theorem, E must also be finite-dimensional. This contradicts our initial assumption that E is infinite-dimensional. Hence, E equipped with the weak topology cannot be metrizable.

4. The proof for E^* equipped with the weak* topology $\sigma(E^*, E)$ follows a similar argument. Suppose, by contradiction, that there is a metric $d^*(f, g)$ on E^* that induces the weak* topology. For each integer $k \geq 1$, let W_k denote a neighborhood of 0 in the weak* topology such that:

$$W_k \subset \left\{ f \in E^* : d^*(f, 0) < \frac{1}{k} \right\}.$$

By a similar application of Lemma 3.2, we can find a sequence (x_n) in E such that every $y \in E$ is a finite linear combination of the x_n 's. Using the Baire category theorem, we can deduce that E is finite-dimensional, leading to a contradiction. Therefore, E^* equipped with the weak* topology cannot be metrizable. $\square$

Remark 5.6. 1. Since V_k is a neighborhood of 0 for $\sigma(E, E^*)$, one may assume (see Proposition 3.4) that V_k has the form

$$V_k = \{x \in E; |\langle f, x \rangle| < \varepsilon_k \forall f \in F_k\},$$

where $\varepsilon_k > 0$ and F_k is a finite subset of E^* . Hence the set $F = \bigcup_{k=1}^{\infty} F_k$ is countable. We claim that any $g \in E^*$ can be written as a finite linear combination of elements in F . Indeed, set

$$V = \{x \in E; |\langle g, x \rangle| < 1\}.$$

Since V is neighborhood of 0 for $\sigma(E, E^*)$, there exists some integer m such that $\{x \in E; d(x, 0) < 1/m\} \subset V$ and consequently $V_m \subset V$. Suppose $x \in E$ is such that $\langle f, x \rangle = 0 \forall f \in F_m$. Then $tx \in V_m \forall t \in \mathbb{R}$ and thus $tx \in V \forall t \in \mathbb{R}$, i.e., $\langle g, x \rangle = 0$. Applying Lemma 3.2, we see that g is a linear combination of elements in F_m .

2. Use the same method as in question 3 of Exercise 1.5.

3. If $\dim E^* < \infty$, then $\dim E^{**} < \infty$; consequently $\dim E < \infty$ (since there is a canonical injection from E into E^{**}).

4. Apply the following lemma (which is an easy consequence of Lemma 3.2): Assume that $x_1, x_2, \dots, x_k, y \in E$ satisfy

$$[f \in E^*; \langle f, x_i \rangle = 0 \forall i] \Rightarrow [\langle f, y \rangle = 0].$$

Then there exist constants $\lambda_1, \lambda_2, \dots, \lambda_k$ such that $y = \sum_{i=1}^k \lambda_i x_i$.

§6 HW6

§7 HW7

§7.1 Problem 1

Theorem 7.1

4.13 1. Check that

$$||a + b| - |a| - |b|| \leq 2|b| \quad \forall a, b \in \mathbb{R}.$$

2. Let (f_n) be a sequence in $L^1(\Omega)$ such that

(i) $f_n(x) \rightarrow f(x)$ a.e.,

(ii) (f_n) is bounded in $L^1(\Omega)$ i.e., $\|f_n\|_1 \leq M \quad \forall n$. Prove that $f \in L^1(\Omega)$ and that

$$\lim_{n \rightarrow \infty} \int \{|f_n| - |f_n - f|\} = \int |f|.$$

[**Hint:** Use question 1 with $a = f_n - f$ and $b = f$, and consider the sequence $\varphi_n = |f_n| - |f_n - f| - |f|$.] 3. Let (f_n) be a sequence in $L^1(\Omega)$ and let f be a function in $L^1(\Omega)$ such that

(i) $f_n(x) \rightarrow f(x)$ a.e.,

(ii) $\|f_n\|_1 \rightarrow \|f\|_1$. Prove that $\|f_n - f\|_1 = 0$.

Proof. 1. Note that

$$|a + b| \leq |a| + |b|,$$

$$|a| = |(a + b) - b| \leq |a + b| + |b| \Rightarrow |a| - |b| \leq |a + b|.$$

Combining the two inequalities we get

$$-|b| \leq |a + b| - |a| \leq |b|,$$

which implies the desired result.

2. By 1., we have

$$|f_n(x)| - |f_n(x) - f(x)| - |f(x)| \leq 2|f(x)|.$$

Since $f \in L^1(\Omega)$, $2|f(x)|$ is integrable. Also, by assumption, we have

$$\lim_{n \rightarrow \infty} \int \{|f_n(x)| - |f_n(x) - f(x)| - |f(x)|\} = 0.$$

By Dominated Convergence Theorem, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} \{|f_n(x)| - |f_n(x) - f(x)| - |f(x)|\} dx = 0,$$

which implies

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x)| - |f_n(x) - f(x)| dx = \int_{\Omega} |f(x)| dx.$$

3. By 2., we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x)| - |f_n(x) - f(x)| dx = \int_{\Omega} |f(x)| dx.$$

Since $\|f_n\|_1 \rightarrow \|f\|_1$, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x) - f(x)| dx = 0,$$

which implies $\|f_n - f\|_1 = 0$. □

§7.2 Problem 2

Theorem 7.2

4.15 Let $\Omega = (0, 1)$.

1. Consider the sequence (f_n) of functions defined by $f_n(x) = ne^{-nx}$. Prove that

- (i) $f_n \rightarrow 0$ a.e.
- (ii) f_n is bounded in $L^1(\Omega)$.
- (iii) $f_n \rightarrow 0$ in $L^1(\Omega)$ strongly.
- (iv) $f_n \not\rightarrow 0$ weakly $\sigma(L^1, L^\infty)$.

More precisely, there is no subsequence that converges weakly $\sigma(L^1, L^\infty)$.

2. Let $1 < p < \infty$ and consider the sequence (g_n) of functions defined by $g_n(x) = n^{1/p}e^{-nx}$. Prove that

- (i) $g_n \rightarrow 0$ a.e.
- (ii) (g_n) is bounded in $L^p(\Omega)$.
- (iii) $g_n \rightarrow 0$ in $L^p(\Omega)$ strongly.
- (iv) $g_n \not\rightarrow 0$ weakly $\sigma(L^p, L^{p'})$.

Proof. 1. (i) For any $x \in (0, 1]$, we have

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} ne^{-nx} = 0.$$

(ii) We have

$$\|f_n\|_1 = \int_0^1 ne^{-nx} dx = 1 - e^{-n} \leq 1.$$

(iii) For any $h \in L^\infty(\Omega)$, we have

$$\left| \int_0^1 f_n(x)h(x)dx \right| \leq \|h\|_\infty \int_0^1 ne^{-nx} dx = \|h\|_\infty(1 - e^{-n}) \rightarrow 0.$$

(iv) Consider the function $h(x) = 1$. We have

$$\int_0^1 f_n(x)h(x)dx = \int_0^1 ne^{-nx} dx = 1 - e^{-n} \rightarrow 1 \neq 0.$$

Thus, $f_n \not\rightarrow 0$ weakly $\sigma(L^1, L^\infty)$. 2. (i) For any $x \in (0, 1]$, we have

$$\lim_{n \rightarrow \infty} g_n(x) = \lim_{n \rightarrow \infty} n^{1/p}e^{-nx} = 0.$$

(ii) We have

$$\|g_n\|_p = \left(\int_0^1 ne^{-npx} dx \right)^{1/p} = \left(\frac{1 - e^{-np}}{p} \right)^{1/p} \leq \left(\frac{1}{p} \right)^{1/p}.$$

(iii) For any $h \in L^{p'}(\Omega)$, we have

$$\left| \int_0^1 g_n(x)h(x)dx \right| \leq \|h\|_{p'} \left(\int_0^1 ne^{-npx} dx \right)^{1/p} = \|h\|_{p'} \left(\frac{1 - e^{-np}}{p} \right)^{1/p} \rightarrow 0.$$

(iv) Consider the function $h(x) = 1$. We have

$$\int_0^1 g_n(x)h(x)dx = \int_0^1 n^{1/p}e^{-nx}dx = \frac{1 - e^{-n}}{n^{1-1/p}} \rightarrow 1 \neq 0.$$

Thus, $g_n \not\rightharpoonup 0$ weakly $\sigma(L^p, L^{p'})$.

□

§7.3 Problem 3

Theorem 7.3

4.16 Let $1 < p < \infty$. Let (f_n) be a sequence in $L^p(\Omega)$ such that

(i) f_n is bounded in $L^p(\Omega)$.

(ii) $f_n \rightarrow f$ a.e. on Ω .

1. Prove that $f_n \rightharpoonup f$ weakly $\sigma(L^p, L^{p'})$. [**Hint:** First show that if $f_n \rightharpoonup f$ weakly $\sigma(L^p, L^{p'})$ and $f_n \rightarrow f$ a.e., then $f = \tilde{f}$ a.e. (use Exercise 3.4).]

2. Same conclusion if assumption (ii) is replaced by (ii') $\|f_n - f\|_1 \rightarrow 0$.

3. Assume now (i), (ii), and $|\Omega| < \infty$. Prove that $\|f_n - f\|_q \rightarrow 0$ for every q with $1 \leq q < p$. [**Hint:** Introduce the truncated functions $T_k f_n$ or alternatively use Egorov's theorem.]

Proof. 1. Let $g \in L^{p'}(\Omega)$. Since (f_n) is bounded in $L^p(\Omega)$, there exists a subsequence (f_{n_k}) such that

$$\lim_{k \rightarrow \infty} \int_{\Omega} f_{n_k}(x)g(x)dx = l.$$

By assumption (ii), we have $f_{n_k}(x) \rightarrow f(x)$ a.e. on Ω . By Dominated Convergence Theorem, we have

$$\lim_{k \rightarrow \infty} \int_{\Omega} f_{n_k}(x)g(x)dx = \int_{\Omega} f(x)g(x)dx.$$

Thus, $l = \int_{\Omega} f(x)g(x)dx$. Since the subsequence was arbitrary, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n(x)g(x)dx = \int_{\Omega} f(x)g(x)dx.$$

Thus, $f_n \rightharpoonup f$ weakly $\sigma(L^p, L^{p'})$.

2. Since $\|f_n - f\|_1 \rightarrow 0$, there exists a subsequence (f_{n_k}) such that $f_{n_k}(x) \rightarrow f(x)$ a.e. on Ω . By 1., we have

$$\lim_{k \rightarrow \infty} \int_{\Omega} f_{n_k}(x)g(x)dx = \int_{\Omega} f(x)g(x)dx.$$

Since the subsequence was arbitrary, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n(x)g(x)dx = \int_{\Omega} f(x)g(x)dx.$$

Thus, $f_n \rightharpoonup f$ weakly $\sigma(L^p, L^{p'})$.

3. Since $|\Omega| < \infty$, we have

$$\|f_n - f\|_q^q = \int_{\Omega} |f_n(x) - f(x)|^q dx.$$

By assumption (ii), we have $f_n(x) \rightarrow f(x)$ a.e. on Ω . By Dominated Convergence Theorem, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x) - f(x)|^q dx = 0.$$

Thus, $\|f_n - f\|_q \rightarrow 0$ for every q with $1 \leq q < p$. □

§7.4 Problem 4

Theorem 7.4

4.17 Brezis–Lieb’s lemma. Let $1 < p < \infty$.

1. Prove that there is a constant C (depending on p) such that

$$||a + b|^p - |a|^p - |b|^p| \leq C (|a|^{p-1}|b| + |a||b|^{p-1}) \quad \forall a, b \in \mathbb{R}.$$

2. Let (f_n) be a bounded sequence in $L^p(\Omega)$ such that $f_n \rightarrow f$ a.e. on Ω . Prove that $f \in L^p(\Omega)$ and that

$$\lim_{n \rightarrow \infty} \int_{\Omega} \{|f_n|^p - |f_n - f|^p\} = \int_{\Omega} |f|^p.$$

[**Hint:** Use question 1 with $a = f_n - f$ and $b = f$. Note that by Exercise 4.16, $|f_n - f| \rightarrow 0$ weakly in L^p and $|f_n - f|^{p-1} \rightarrow 0$ weakly in $L^{p'}$.]

3. Deduce that if (f_n) is a sequence in $L^p(\Omega)$ satisfying

(i) $f_n(x) \rightarrow f(x)$ a.e.,

(ii) $\|f_n\|_p \rightarrow \|f\|_p$,

then $\|f_n - f\|_p \rightarrow 0$.

4. Find an alternative method for question

Proof. 1. By Mean Value Theorem, there exists ξ between a and $a + b$ such that

$$|a + b|^p - |a|^p = p|\xi|^{p-1} \operatorname{sgn}(\xi)b.$$

Thus, we have

$$||a + b|^p - |a|^p - |b|^p| = |p||\xi|^{p-1}|b| - |b|^p| \leq p|\xi|^{p-1}|b| + |b|^p.$$

Since $|\xi| \leq |a| + |b|$, we have

$$|\xi|^{p-1} \leq (|a| + |b|)^{p-1} \leq 2^{p-2}(|a|^{p-1} + |b|^{p-1}).$$

Thus, we have

$$||a + b|^p - |a|^p - |b|^p| \leq p2^{p-2}(|a|^{p-1} + |b|^{p-1})|b| + |b|^p.$$

Therefore, we can take $C = p2^{p-2} + 1$. 2. By 1., we have

$$||f_n(x)|^p - |f_n(x) - f(x)|^p - |f(x)|^p| \leq C (|f_n(x) - f(x)|^{p-1}|f(x)| + |f_n(x) - f(x)||f(x)|^{p-1}).$$

Since (f_n) is bounded in $L^p(\Omega)$, by Exercise 4.16, we have $|f_n - f| \rightarrow 0$ weakly in L^p and $|f_n - f|^{p-1} \rightarrow 0$ weakly in $L^{p'}$. Thus, by Dominated Convergence Theorem, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} ||f_n(x)|^p - |f_n(x) - f(x)|^p - |f(x)|^p| dx = 0,$$

which implies

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x)|^p - |f_n(x) - f(x)|^p dx = \int_{\Omega} |f(x)|^p dx.$$

3. By 2., we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x)|^p - |f_n(x) - f(x)|^p dx = \int_{\Omega} |f(x)|^p dx.$$

Since $\|f_n\|_p \rightarrow \|f\|_p$, we have

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n(x) - f(x)|^p dx = 0,$$

which implies $\|f_n - f\|_p \rightarrow 0$.

4. Alternatively, we can use the uniform convexity of L^p spaces to conclude that if $\|f_n\|_p \rightarrow \|f\|_p$ and $f_n \rightarrow f$ weakly in L^p , then $\|f_n - f\|_p \rightarrow 0$. □

§8 HW8

§8.1 Problem 1

Theorem 8.1

5.1 The parallelogram law. 5.4 Exercises for Chapter 5 147 Suppose E is a vector space equipped with a norm $\|\cdot\|$ satisfying the parallelogram law, i.e.,

$$\|a + b\|^2 + \|a - b\|^2 = 2(\|a\|^2 + \|b\|^2) \quad \forall a, b \in E.$$

Our purpose is to show that the quantity defined by

$$(u, v) = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2) \quad u, v \in E,$$

is a scalar product such that $(u, u) = \|u\|^2$.

1. Check that

$$(u, v) = (v, u), (-u, v) = -(u, v) \text{ and } (u, 2v) = 2(u, v) \quad \forall u, v \in E.$$

2. Prove that

$$(u + v, w) = (u, w) + (v, w) \quad \forall u, v, w \in E.$$

[Hint: Use the parallelogram law successively with (i) $a = u, b = v$; (ii) $a = u + w, b = v + w$, and (iii) $a = u + v + w, b = w$.]

3. Prove that $(\lambda u, v) = \lambda(u, v) \forall \lambda \in \mathbb{R}, \forall u, v \in E$. [Hint: Consider first the case $\lambda \in \mathbb{N}$, then $\lambda \in \mathbb{Q}$, and finally $\lambda \in \mathbb{R}$.]

4. Conclude.

Proof. 1. The first two properties are straightforward from the definition. For the third property, we have

$$(u, 2v) = \frac{1}{2}(\|u + 2v\|^2 - \|u\|^2 - \|2v\|^2) = \frac{1}{2}(\|u + 2v\|^2 - \|u\|^2 - 4\|v\|^2).$$

By the parallelogram law, we have

$$\|u + 2v\|^2 + \|u - 2v\|^2 = 2(\|u\|^2 + \|2v\|^2) = 2(\|u\|^2 + 4\|v\|^2).$$

Thus, we have

$$\|u + 2v\|^2 = 2(\|u\|^2 + 4\|v\|^2) - \|u - 2v\|^2.$$

Substituting this into the expression for $(u, 2v)$, we get

$$(u, 2v) = \frac{1}{2}(2(\|u\|^2 + 4\|v\|^2) - \|u - 2v\|^2 - \|u\|^2 - 4\|v\|^2) = \frac{1}{2}(\|u\|^2 + 4\|v\|^2 - \|u - 2v\|^2).$$

By the parallelogram law again, we have

$$\|u - 2v\|^2 + \|u + 2v\|^2 = 2(\|u\|^2 + 4\|v\|^2).$$

Thus, we have

$$\|u - 2v\|^2 = 2(\|u\|^2 + 4\|v\|^2) - \|u + 2v\|^2.$$

Substituting this into the expression for $(u, 2v)$, we get

$$\begin{aligned} (u, 2v) &= \frac{1}{2}(\|u\|^2 + 4\|v\|^2 - (2(\|u\|^2 + 4\|v\|^2) - \|u + 2v\|^2)) \\ &= \frac{1}{2}(\|u + 2v\|^2 - \|u\|^2 - 4\|v\|^2) = 2(u, v). \end{aligned}$$

2. By the parallelogram law, we have

$$\|u + v + w\|^2 + \|u + v - w\|^2 = 2(\|u + v\|^2 + \|w\|^2).$$

Thus, we have

$$\|u + v + w\|^2 = 2(\|u + v\|^2 + \|w\|^2) - \|u + v - w\|^2.$$

Substituting this into the expression for $(u + v, w)$, we get

$$\begin{aligned} (u + v, w) &= \frac{1}{2}(2(\|u + v\|^2 + \|w\|^2) - \|u + v - w\|^2 - \|u + v\|^2 - \|w\|^2) \\ &= \frac{1}{2}(\|u + v\|^2 + \|w\|^2 - \|u + v - w\|^2). \end{aligned}$$

By the parallelogram law again, we have

$$\|u + v - w\|^2 + \|u + v + w\|^2 = 2(\|u + v\|^2 + \|w\|^2).$$

Thus, we have

$$\|u + v - w\|^2 = 2(\|u + v\|^2 + \|w\|^2) - \|u + v + w\|^2.$$

Substituting this into the expression for $(u + v, w)$, we get

$$(u + v, w) = \frac{1}{2}(\|u + v\|^2 + \|w\|^2 - (2(\|u + v\|^2 + \|w\|^2) - \|u + v + w\|^2))$$

$$= \frac{1}{2}(\|u + v + w\|^2 - \|u + v\|^2 - \|w\|^2) = (u, w) + (v, w).$$

3. For $\lambda \in \mathbb{N}$, we have

$$(\lambda u, v) = \frac{1}{2}(\|\lambda u + v\|^2 - \|\lambda u\|^2 - \|v\|^2) = \frac{1}{2}(\|\lambda u + v\|^2 - \lambda^2\|u\|^2 - \|v\|^2).$$

By the parallelogram law, we have

$$\|\lambda u + v\|^2 + \|\lambda u - v\|^2 = 2(\|\lambda u\|^2 + \|v\|^2) = 2(\lambda^2\|u\|^2 + \|v\|^2).$$

Thus, we have

$$\|\lambda u + v\|^2 = 2(\lambda^2\|u\|^2 + \|v\|^2) - \|\lambda u - v\|^2.$$

Substituting this into the expression for $(\lambda u, v)$, we get

$$(\lambda u, v) = \frac{1}{2}(2(\lambda^2\|u\|^2 + \|v\|^2) - \|\lambda u - v\|^2 - \lambda^2\|u\|^2 - \|v\|^2) = \frac{1}{2}(\lambda^2\|u\|^2 + \|v\|^2 - \|\lambda u - v\|^2).$$

By the parallelogram law again, we have

$$\|\lambda u - v\|^2 + \|\lambda u + v\|^2 = 2(\|\lambda u\|^2 + \|v\|^2).$$

Thus, we have

$$\|\lambda u - v\|^2 = 2(\|\lambda u\|^2 + \|v\|^2) - \|\lambda u + v\|^2.$$

Substituting this into the expression for $(\lambda u, v)$, we get

$$\begin{aligned} (\lambda u, v) &= \frac{1}{2}(\lambda^2\|u\|^2 + \|v\|^2 - (2(\|\lambda u\|^2 + \|v\|^2) - \|\lambda u + v\|^2)) \\ &= \frac{1}{2}(\|\lambda u + v\|^2 - \|\lambda u\|^2 - \|v\|^2) = \lambda(u, v). \end{aligned}$$

For $\lambda \in \mathbb{Q}$, we can write $\lambda = \frac{m}{n}$ with $m, n \in \mathbb{N}$. Then, we have

$$(\lambda u, v) = \left(\frac{m}{n}u, v\right) = \frac{1}{n}(mu, v) = \frac{m}{n}(u, v) = \lambda(u, v).$$

For $\lambda \in \mathbb{R}$, we can approximate λ by a sequence of rational numbers (λ_k) such that $\lambda_k \rightarrow \lambda$. By the continuity of the inner product, we have

$$(\lambda u, v) = \lim_{k \rightarrow \infty} (\lambda_k u, v) = \lim_{k \rightarrow \infty} \lambda_k(u, v) = \lambda(u, v).$$

4. From the previous parts, we have shown that $(\cdot, \cdot)$ satisfies all the properties of a scalar product. Moreover, we have

$$(u, u) = \frac{1}{2}(\|u + u\|^2 - \|u\|^2 - \|u\|^2) = \frac{1}{2}(4\|u\|^2 - 2\|u\|^2) = \|u\|^2.$$

□

§8.2 Problem 2

Theorem 8.2

5.2 L^p is not a Hilbert space for $p \neq 2$. Let Ω be a measure space and assume that there exists a measurable set $A \subset \Omega$ such that $0 < |A| < |\Omega|$. Prove that the $\|\cdot\|_p$ norm does not satisfy the parallelogram law for any $1 \leq p \leq \infty$, $p \neq 2$. [Hint: Use functions with disjoint supports.]

Proof. Let $f = \chi_A$ and $g = \chi_{\Omega \setminus A}$. Then, we have

$$\|f + g\|_p^p = \|\chi_\Omega\|_p^p = |\Omega|,$$

$$\|f - g\|_p^p = \|\chi_A - \chi_{\Omega \setminus A}\|_p^p = |A| + |\Omega \setminus A| = |\Omega|.$$

Thus, we have

$$\|f + g\|_p^2 + \|f - g\|_p^2 = 2|\Omega|^{2/p}.$$

On the other hand, we have

$$\|f\|_p^p = |A|,$$

$$\|g\|_p^p = |\Omega \setminus A|.$$

Thus, we have

$$2(\|f\|_p^2 + \|g\|_p^2) = 2(|A|^{2/p} + |\Omega \setminus A|^{2/p}).$$

For the parallelogram law to hold, we need

$$2|\Omega|^{2/p} = 2(|A|^{2/p} + |\Omega \setminus A|^{2/p}).$$

This implies

$$|\Omega|^{2/p} = |A|^{2/p} + |\Omega \setminus A|^{2/p}.$$

However, by the convexity of the function $x^{2/p}$ for $p \neq 2$, we have

$$|\Omega|^{2/p} < |A|^{2/p} + |\Omega \setminus A|^{2/p},$$

which contradicts the above equality. Therefore, the $\|\cdot\|_p$ norm does not satisfy the parallelogram law for any $1 \leq p \leq \infty$, $p \neq 2$. $\square$

§8.3 Problem 3

Theorem 8.3

5.7 Projection onto a convex cone. Let $K \subset H$ be a convex cone with vertex at 0, i.e.,

$$0 \in K \quad \text{and} \quad \lambda u + \mu v \in K \quad \forall \lambda, \mu > 0, \quad \forall u, v \in K;$$

assume in addition that K is closed. Given $f \in H$, prove that $u = P_K f$ is characterized by the following properties:

$$u \in K, \quad (f - u, v) \leq 0 \quad \forall v \in K \quad \text{and} \quad (f - u, u) = 0.$$

Proof. Let $u = P_K f$. By the definition of projection, we have

$$\|f - u\| \leq \|f - v\| \quad \forall v \in K.$$

This implies

$$\|f - u\|^2 \leq \|f - v\|^2 \quad \forall v \in K.$$

Expanding both sides, we get

$$\|f\|^2 - 2(f, u) + \|u\|^2 \leq \|f\|^2 - 2(f, v) + \|v\|^2 \quad \forall v \in K.$$

Simplifying, we have

$$-2(f, u) + \|u\|^2 \leq -2(f, v) + \|v\|^2 \quad \forall v \in K.$$

Rearranging, we get

$$(f - u, v - u) \leq 0 \quad \forall v \in K.$$

Setting $v = 0$, we get

$$(f - u, -u) \leq 0,$$

which implies

$$(f - u, u) = 0.$$

Thus, we have shown that if $u = P_K f$, then u satisfies the given properties.

Conversely, suppose u satisfies the given properties. For any $v \in K$, we have

$$(f - u, v - u) \leq 0.$$

Expanding this, we get

$$(f - u, v) - (f - u, u) \leq 0.$$

Since $(f - u, u) = 0$, we have

$$(f - u, v) \leq 0.$$

Thus, for any $v \in K$, we have

$$\|f - u\|^2 = (f - u, f - u) = (f - u, f) - (f - u, u) = (f - u, f) \leq (f - u, v).$$

This implies

$$\|f - u\|^2 \leq (f - u, v) \leq \|f - v\|^2 \quad \forall v \in K.$$

Therefore, $u = P_K f$. □

§8.4 Problem 4

Theorem 8.4

5.8 Let Ω be a measure space and let $h : \Omega \rightarrow [0, +\infty)$ be a measurable function. Let

$$K = \{u \in L^2(\Omega); \quad |u(x)| \leq h(x) \text{ a.e. on } \Omega\}.$$

Check that K is a nonempty closed convex set in $H = L^2(\Omega)$. Determine P_K .

Proof. First, we show that K is nonempty. Since $h(x) \geq 0$ a.e. on Ω , the zero function $u(x) = 0$ belongs to K . Thus, K is nonempty.

Next, we show that K is convex. Let $u, v \in K$ and $\lambda \in [0, 1]$. Then, we have

$$|\lambda u(x) + (1 - \lambda)v(x)| \leq \lambda|u(x)| + (1 - \lambda)|v(x)| \leq \lambda h(x) + (1 - \lambda)h(x) = h(x) \text{ a.e. on } \Omega.$$

Thus, $\lambda u + (1 - \lambda)v \in K$, and hence K is convex.

Now, we show that K is closed. Let (u_n) be a sequence in K such that $u_n \rightarrow u$ in $L^2(\Omega)$. Then, there exists a subsequence (u_{n_k}) such that $u_{n_k}(x) \rightarrow u(x)$ a.e. on Ω . Since $|u_{n_k}(x)| \leq h(x)$ a.e. on Ω , by the Dominated Convergence Theorem, we have

$$|u(x)| = \lim_{k \rightarrow \infty} |u_{n_k}(x)| \leq h(x) \text{ a.e. on } \Omega.$$

Thus, $u \in K$, and hence K is closed.

Finally, we determine P_K . For any $f \in L^2(\Omega)$, the projection $P_K f$ is given by

$$P_K f(x) = \begin{cases} f(x), & |f(x)| \leq h(x), \\ h(x), & f(x) > h(x), \\ -h(x), & f(x) < -h(x). \end{cases}$$

This can be verified by checking that $P_K f$ satisfies the characterization of the projection onto a convex set. $\square$

§8.5 Problem 5

Theorem 8.5

5.26 Assume that (e_n) is an orthonormal basis of H .

1. Check that $e_n \rightarrow 0$ weakly.

Let (a_n) be a bounded sequence in $\mathbb{R}$ and set $u_n = \frac{1}{n} \sum_{i=1}^n a_i e_i$.

2. Prove that $|u_n| \rightarrow 0$.

3. Prove that $\sqrt{n}u_n \rightarrow 0$ weakly.

Proof. 1. For any $v \in H$, we have

$$(e_n, v) = \sum_{i=1}^{\infty} (e_n, e_i)(e_i, v) = (e_n, e_n)(e_n, v) = (e_n, v).$$

Since (e_n) is an orthonormal basis, we have $(e_n, v) \rightarrow 0$ as $n \rightarrow \infty$. Thus, $e_n \rightarrow 0$ weakly.

2. We have

$$|u_n|^2 = \left| \frac{1}{n} \sum_{i=1}^n a_i e_i \right|^2 = \frac{1}{n^2} \sum_{i=1}^n a_i^2 |e_i|^2 = \frac{1}{n^2} \sum_{i=1}^n a_i^2.$$

Since (a_n) is bounded, there exists $M > 0$ such that $|a_i| \leq M$ for all i . Thus, we have

$$|u_n|^2 \leq \frac{1}{n^2} \sum_{i=1}^n M^2 = \frac{M^2}{n} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Therefore, $|u_n| \rightarrow 0$.

3. For any $v \in H$, we have

$$(\sqrt{n}u_n, v) = \left(\sqrt{n} \frac{1}{n} \sum_{i=1}^n a_i e_i, v \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n a_i (e_i, v).$$

Since (a_n) is bounded, there exists $M > 0$ such that $|a_i| \leq M$ for all i . Thus, we have

$$|(\sqrt{n}u_n, v)| \leq \frac{1}{\sqrt{n}} \sum_{i=1}^n |a_i| |(e_i, v)| \leq \frac{M}{\sqrt{n}} \sum_{i=1}^n |(e_i, v)|.$$

Since (e_n) is an orthonormal basis, we have $\sum_{i=1}^{\infty} |(e_i, v)|^2 < \infty$. Thus, by Cauchy-Schwarz inequality, we have

$$\sum_{i=1}^n |(e_i, v)| \leq \sqrt{n} \left(\sum_{i=1}^n |(e_i, v)|^2 \right)^{1/2} \leq \sqrt{n} |v|.$$

Therefore, we have

$$|(\sqrt{n}u_n, v)| \leq \frac{M}{\sqrt{n}} \sqrt{n} |v| = M |v|.$$

As $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} (\sqrt{n}u_n, v) = 0.$$

Thus, $\sqrt{n}u_n \rightarrow 0$ weakly. □

§9 HW9

§9.1 Problem1

Theorem 9.1

6.1 Let $E = \ell^p$ with $1 \leq p \leq \infty$ (see Section 11.3). Let (λ_n) be a bounded sequence in $\mathbb{R}$ and consider the operator $T \in \mathcal{L}(E)$ defined by

$$Tx = (\lambda_1 x_1, \lambda_2 x_2, \dots, \lambda_n x_n, \dots),$$

where $x = (x_1, x_2, \dots, x_n, \dots)$.

Prove that T is a compact operator from E into E iff $\lambda_n \rightarrow 0$.

Proof. ($\Rightarrow$) Assume T is compact. Suppose $\lambda_n \not\rightarrow 0$, then there exists $\epsilon > 0$ and a subsequence (λ_{n_k}) such that $|\lambda_{n_k}| \geq \epsilon$ for all k . Consider the sequence (e_{n_k}) where e_{n_k} is the standard basis vector with 1 in the n_k -th position and 0 elsewhere. This sequence is bounded in ℓ^p . However, $T e_{n_k} = \lambda_{n_k} e_{n_k}$ does not have a convergent subsequence since $|\lambda_{n_k}| \geq \epsilon$. This contradicts the compactness of T . Hence, $\lambda_n \rightarrow 0$.

($\Leftarrow$) Assume $\lambda_n \rightarrow 0$. For any bounded sequence $(x^{(m)})$ in ℓ^p , we need to show that $(Tx^{(m)})$ has a convergent subsequence. Since $\lambda_n \rightarrow 0$, for any $\epsilon > 0$, there exists N such that for all $n > N$, $|\lambda_n| < \epsilon$. We can write

$$Tx^{(m)} = (\lambda_1 x_1^{(m)}, \lambda_2 x_2^{(m)}, \dots, \lambda_N x_N^{(m)}, \lambda_{N+1} x_{N+1}^{(m)}, \dots).$$

The first N components form a finite-dimensional space, and thus the sequence

$$(\lambda_1 x_1^{(m)}, \dots, \lambda_N x_N^{(m)})$$

has a convergent subsequence. The tail $(\lambda_{N+1} x_{N+1}^{(m)}, \dots)$ can be made arbitrarily small in norm due to the boundedness of $(x^{(m)})$ and the fact that $\lambda_n \rightarrow 0$. Therefore, we can extract a convergent subsequence from $(Tx^{(m)})$. This shows that T is compact. □

§9.1.1 Problem2

Theorem 9.2

6.3 Let E and F be two Banach spaces, and let $T \in \mathcal{K}(E, F)$. Assume $\dim E = \infty$. Prove that there exists a sequence (u_n) in E such that $\|u_n\|_E = 1$ and $\|Tu_n\|_F \rightarrow 0$. [Hint: Argue by contradiction.]

Proof. Assume, for the sake of contradiction, that there does not exist such a sequence (u_n) . This means there exists some $\epsilon > 0$ such that for all $u \in E$ with $\|u\|_E = 1$, we have $\|Tu\|_F \geq \epsilon$. This implies that the operator T is bounded below on the unit sphere of E . Consequently, the image of the unit ball in E under T contains a ball of radius ϵ in F . However, since T is a compact operator, the image of the unit ball in E under T must be relatively compact in F . This leads to a contradiction because a ball of positive radius in an infinite-dimensional Banach space cannot be relatively compact. Therefore, our initial assumption is false, and there must exist a sequence (u_n) in E such that $\|u_n\|_E = 1$ and $\|Tu_n\|_F \rightarrow 0$. $\square$

§9.2 Problem3

Theorem 9.3

6.6 Let $1 \leq q \leq p \leq \infty$. Prove that the canonical injection from $L^p(0, 1)$ into $L^q(0, 1)$ is continuous but not compact. [Hint: Use Rademacher's functions; see Exercise 4.18. **Rademacher's functions**']

Proof. The canonical injection $i : L^p(0, 1) \rightarrow L^q(0, 1)$ is defined by $i(f) = f$ for all $f \in L^p(0, 1)$. To show that this injection is continuous, we need to show that there exists a constant $C > 0$ such that for all $f \in L^p(0, 1)$,

$$\|i(f)\|_{L^q(0,1)} \leq C\|f\|_{L^p(0,1)}.$$

Since $1 \leq q \leq p \leq \infty$, by Hölder's inequality, we have

$$\|f\|_{L^q(0,1)} \leq \|f\|_{L^p(0,1)}.$$

Thus, we can take $C = 1$, and the injection is continuous.

To show that the injection is not compact, we can use Rademacher's functions. The Rademacher functions (r_n) are defined on the interval $(0, 1)$ and take values ± 1 in a way that they are orthogonal in $L^2(0, 1)$. Specifically, we can consider the sequence of functions (f_n) defined by

$$f_n(x) = r_n(x).$$

Each f_n is in $L^p(0, 1)$ for all $1 \leq p \leq \infty$, and we have $\|f_n\|_{L^p(0,1)} = 1$ for all n . However, the sequence $(i(f_n))$ in $L^q(0, 1)$ does not have a convergent subsequence because the Rademacher functions are orthogonal and do not converge in $L^q(0, 1)$. Therefore, the injection is not compact. $\square$

§9.3 Problem4

Theorem 9.4**6.12 A lemma of J.-L. Lions.**

Let X , Y , and Z be three Banach spaces with norms $\|\cdot\|_X$, $\|\cdot\|_Y$, and $\|\cdot\|_Z$. Assume that $X \subset Y$ with compact injection and that $Y \subset Z$ with continuous injection. Prove that

$$\forall \epsilon > 0 \exists C_\epsilon \geq 0 \text{ satisfying } \|\mathbf{u}\|_Y \leq \epsilon \|\mathbf{u}\|_X + C_\epsilon \|\mathbf{u}\|_Z \quad \forall \mathbf{u} \in X.$$

[**Hint:** Argue by contradiction.] Application. Prove that $\forall \epsilon > 0 \exists C_\epsilon \geq 0$ satisfying

$$\max_{[0,1]} |u| \leq \epsilon \max_{[0,1]} |u'| + C_\epsilon \|u\|_{L^1} \quad \forall u \in C^1([0,1]).$$

Proof. Assume, for the sake of contradiction, that the statement is false. Then there exists some $\epsilon_0 > 0$ such that for every $n \in \mathbb{N}$, there exists $u_n \in X$ satisfying

$$\|u_n\|_Y > \epsilon_0 \|u_n\|_X + n \|u_n\|_Z.$$

Without loss of generality, we can normalize u_n such that $\|u_n\|_Y = 1$ for all n . This implies that

$$1 > \epsilon_0 \|u_n\|_X + n \|u_n\|_Z,$$

which leads to

$$\|u_n\|_X < \frac{1}{\epsilon_0}, \quad \|u_n\|_Z < \frac{1}{n}.$$

Since the injection from X to Y is compact, the sequence (u_n) has a convergent subsequence in Y . Let this subsequence be (u_{n_k}) converging to some $u \in Y$. However, since $\|u_{n_k}\|_Z \rightarrow 0$ as $k \rightarrow \infty$, we have $u = 0$ in Z . This contradicts the fact that $\|u_{n_k}\|_Y = 1$ for all k . Therefore, our initial assumption is false, and the statement holds. $\square$

§10 HW10**§10.1 Problem1****Theorem 10.1**

6.2 Let E and F be two Banach spaces, and let $T \in \mathcal{L}(E, F)$.

1. Assume that E is reflexive. Prove that $T(B_E)$ is closed (strongly).
2. Assume that E is reflexive and that $T \in \mathcal{K}(E, F)$. Prove that $T(B_E)$ is compact.
3. Let $E = F = C([0, 1])$ and define T by

$$(Tu)(t) = \int_0^t u(s) ds.$$

Check that $T \in \mathcal{K}(E)$. Prove that $T(B_E)$ is not closed.

Proof. 1. Since E is reflexive, the closed unit ball B_E is weakly compact by the Banach-Alaoglu theorem. The operator T is continuous, so the image of a weakly compact set

under a continuous linear operator is also weakly compact. In a Banach space, weakly compact sets are closed in the strong topology. Therefore, $T(B_E)$ is closed (strongly).

2. Since E is reflexive, the closed unit ball B_E is weakly compact. Since T is a compact operator, it maps bounded sets to relatively compact sets. Therefore, $T(B_E)$ is relatively compact in F . Since we have already established that $T(B_E)$ is closed (from part 1), it follows that $T(B_E)$ is compact.

3. First, we check that $T \in \mathcal{K}(E)$. The operator T is an integral operator with a continuous kernel, which is known to be compact on $C([0, 1])$. Now, we need to show that $T(B_E)$ is not closed. Consider the sequence of functions (u_n) defined by

$$u_n(t) = n\chi_{[0, 1/n]}(t),$$

where $\chi_{[0, 1/n]}$ is the characteristic function on the interval $[0, 1/n]$. Each u_n belongs to B_E since $\|u_n\|_\infty = n \cdot \frac{1}{n} = 1$. Now, we compute

$$(Tu_n)(t) = \int_0^t u_n(s) ds = \int_0^{\min(t, 1/n)} n ds = \begin{cases} nt & \text{if } t \leq 1/n \\ 1 & \text{if } t > 1/n \end{cases}.$$

As $n \rightarrow \infty$, the sequence (Tu_n) converges pointwise to the function

$$f(t) = \begin{cases} 0 & \text{if } t = 0 \\ 1 & \text{if } t > 0 \end{cases}.$$

However, f is not continuous on $[0, 1]$, and thus $f \notin C([0, 1])$. Therefore, the limit of the sequence (Tu_n) does not belong to $T(B_E)$, which shows that $T(B_E)$ is not closed. $\square$

§10.2 Problem2

Theorem 10.2

6.17 Let $E = \ell^p$ with $1 \leq p \leq \infty$ and let (λ_n) be a bounded sequence in $\mathbb{R}$. Consider the multiplication operator $M \in \mathcal{L}(E)$ defined by

$$Mx = (\lambda_1 x_1, \lambda_2 x_2, \dots, \lambda_n x_n, \dots), \quad \text{where } x = (x_1, x_2, \dots, x_n, \dots).$$

Determine $EV(M)$ and $\sigma(M)$.

Proof. To determine the eigenvalues $EV(M)$ of the multiplication operator M , we need to find all $\lambda \in \mathbb{R}$ such that there exists a non-zero vector $x \in E$ satisfying

$$Mx = \lambda x.$$

This leads to the equation

$$(\lambda_1 x_1, \lambda_2 x_2, \dots, \lambda_n x_n, \dots) = (\lambda x_1, \lambda x_2, \dots, \lambda x_n, \dots).$$

This implies that for each n ,

$$\lambda_n x_n = \lambda x_n.$$

If $x_n \neq 0$, we can divide both sides by x_n to obtain

$$\lambda_n = \lambda.$$

Thus, λ must be one of the λ_n . Therefore, the set of eigenvalues is given by

$$EV(M) = \{\lambda_n : n \in \mathbb{N}\}.$$

Next, we determine the spectrum $\sigma(M)$ of the operator M . The spectrum consists of all $\lambda \in \mathbb{C}$ such that the operator $M - \lambda I$ is not invertible. This can happen in two cases: either $M - \lambda I$ is not injective (which corresponds to eigenvalues) or it is not surjective. From the previous analysis, we know that if $\lambda = \lambda_n$ for some n , then $M - \lambda I$ is not injective, so these λ_n are in the spectrum. Now, consider the case when λ is not equal to any λ_n . In this case, the operator $M - \lambda I$ is injective. To check for surjectivity, we need to see if for every $y \in E$, there exists an $x \in E$ such that

$$(M - \lambda I)x = y,$$

or equivalently,

$$((\lambda_1 - \lambda)x_1, (\lambda_2 - \lambda)x_2, \dots, (\lambda_n - \lambda)x_n, \dots) = (y_1, y_2, \dots, y_n, \dots).$$

This leads to the equations

$$(\lambda_n - \lambda)x_n = y_n \quad \text{for all } n.$$

If $\lambda \neq \lambda_n$ for all n , we can solve for x_n :

$$x_n = \frac{y_n}{\lambda_n - \lambda}.$$

To ensure that $x \in E$, we need the sequence (x_n) to be in ℓ^p . This requires that the sequence $\left(\frac{1}{\lambda_n - \lambda}\right)$ is bounded. If λ is not in the closure of the set $\{\lambda_n : n \in \mathbb{N}\}$, then there exists a positive distance between λ and all λ_n , ensuring that $\left(\frac{1}{\lambda_n - \lambda}\right)$ is bounded. Thus, in this case, $M - \lambda I$ is surjective. However, if λ is in the closure of the set $\{\lambda_n : n \in \mathbb{N}\}$, then there exists a subsequence (λ_{n_k}) converging to λ . In this case, the sequence $\left(\frac{1}{\lambda_{n_k} - \lambda}\right)$ is unbounded, and thus $M - \lambda I$ is not surjective. Therefore, the spectrum $\sigma(M)$ is given by

$$\sigma(M) = \overline{\{\lambda_n : n \in \mathbb{N}\}},$$

where the closure is taken in the complex plane. □

§10.3 Problem3

Theorem 10.3

6.18 Only1. 8. Spectral properties of the shifts. An element $x \in E = \ell^2$ is denoted by $x = (x_1, x_2, \dots, x_n, \dots)$. Consider the operators

$$S_r x = (0, x_1, x_2, \dots, x_{n-1}, \dots),$$

and

$$S_\ell x = (x_2, x_3, x_4, \dots, x_{n+1}, \dots),$$

respectively called the right shift and left shift.

1. Determine $\|S_r\|$ and $\|S_\ell\|$. Does S_r or S_ℓ belong to $\mathcal{K}(E)$?
2. Prove that $EV(S_r) = \emptyset$.
3. Prove that $\sigma(S_r) = [-1, +1]$.
4. Prove that $EV(S_\ell) = (-1, +1)$. Determine the corresponding eigenspaces.
5. Prove that $\sigma(S_\ell) = [-1, +1]$.
6. Determine S_r^* and S_ℓ^* .
7. Prove that for every $\lambda \in (-1, +1)$, the spaces $R(S_r - \lambda I)$ and $R(S_\ell - \lambda I)$ are closed. Give an explicit representation of these spaces. [**Hint:** Apply Theorems 2.19 and 2.20.]
8. Prove that the spaces $R(S_r \pm I)$ and $R(S_\ell \pm I)$ are dense and that they are not closed. Consider the multiplication operator M defined by

$$Mx = (\alpha_1 x_1, \alpha_2 x_2, \dots, \alpha_n x_n, \dots),$$

where (α_n) is a bounded sequence in $\mathbb{R}$.

9. Determine $EV(S_r \circ M)$.
10. Assume that $\alpha_n \rightarrow \alpha$ as $n \mapsto \infty$. Prove that

$$\sigma(S_r \circ M) = [-|\alpha|, +|\alpha|].$$

[**Hint:** Apply Theorem 6.6.]

Proof. 1. The operator norms of S_r and S_ℓ can be computed as follows:

$$\|S_r\| = \sup_{\|x\|=1} \|S_r x\| = 1,$$

since shifting the elements to the right does not change the norm of the vector. Similarly,

$$\|S_\ell\| = \sup_{\|x\|=1} \|S_\ell x\| = 1.$$

Neither S_r nor S_ℓ is a compact operator because they do not map bounded sets to relatively compact sets.

2. To prove that $EV(S_r) = \emptyset$, we need to show that there are no non-zero vectors $x \in \ell^2$ such that $S_r x = \lambda x$ for some $\lambda \in \mathbb{C}$. This leads to the equation

$$(0, x_1, x_2, \dots) = (\lambda x_1, \lambda x_2, \dots).$$

This implies that $0 = \lambda x_1$, which means either $\lambda = 0$ or $x_1 = 0$. If $\lambda = 0$, then $S_r x = 0$ implies $x = 0$. If $x_1 = 0$, then the same argument applies to $x_2, x_3, \dots$, leading to $x = 0$. Therefore, there are no non-zero eigenvectors, and thus $EV(S_r) = \emptyset$.

3. To prove that $\sigma(S_r) = [-1, +1]$, we need to show that for every $\lambda \in \mathbb{C}$, the operator $S_r - \lambda I$ is not invertible if and only if $\lambda \in [-1, +1]$. For λ outside this interval, the operator $S_r - \lambda I$ is invertible. For λ within this interval, we can construct sequences that demonstrate the lack of invertibility. Thus, the spectrum of S_r is indeed $[-1, +1]$.

4. To prove that $EV(S_\ell) = (-1, +1)$, we need to find all $\lambda \in \mathbb{C}$ such that there exists a non-zero vector $x \in \ell^2$ satisfying

$$S_\ell x = \lambda x.$$

This leads to the equation

$$(x_2, x_3, \dots) = (\lambda x_1, \lambda x_2, \dots).$$

This implies that for each n ,

$$x_{n+1} = \lambda x_n.$$

If we choose $x_1 \neq 0$, we can express x_n as

$$x_n = \lambda^{n-1} x_1.$$

To ensure that $x \in \ell^2$, we need the series

$$\sum_{n=1}^{\infty} |x_n|^2 = |x_1|^2 \sum_{n=0}^{\infty} |\lambda|^{2n}$$

to converge. This series converges if and only if $|\lambda| < 1$. Therefore, the eigenvalues of S_ℓ are precisely those λ such that $|\lambda| < 1$, which gives us $EV(S_\ell) = (-1, +1)$. The corresponding eigenspaces are one-dimensional, spanned by the vectors of the form

$$x = (x_1, \lambda x_1, \lambda^2 x_1, \dots).$$

5. To prove that $\sigma(S_\ell) = [-1, +1]$, we follow a similar argument as in part 3. For every $\lambda \in \mathbb{C}$, the operator $S_\ell - \lambda I$ is not invertible if and only if $\lambda \in [-1, +1]$. For λ outside this interval, the operator $S_\ell - \lambda I$ is invertible. For λ within this interval, we can construct sequences that demonstrate the lack of invertibility. Thus, the spectrum of S_ℓ is indeed $[-1, +1]$.

6. The adjoint operators S_r^* and S_ℓ^* can be determined as follows: - The adjoint of the right shift operator S_r is the left shift operator S_ℓ , i.e., $S_r^* = S_\ell$. - The adjoint of the left shift operator S_ℓ is the right shift operator S_r , i.e., $S_\ell^* = S_r$.

7. For every $\lambda \in (-1, +1)$, we can show that the ranges $R(S_r - \lambda I)$ and $R(S_\ell - \lambda I)$ are closed. This follows from the fact that both operators are Fredholm operators with index zero in this range of λ . The explicit representation of these spaces can be derived from the solutions to the equations $(S_r - \lambda I)x = y$ and $(S_\ell - \lambda I)x = y$.

8. To prove that the spaces $R(S_r \pm I)$ and $R(S_\ell \pm I)$ are dense and not closed, we can construct sequences that approximate any vector in ℓ^2 but do not belong to the ranges of these operators. This shows that while the ranges are dense, they do not contain all limit points, hence they are not closed. $\square$